

Choose the most appropriate option (a, b, c or d).

- Q 1. If $f(x) = x^2 + \lambda x + \mu$ be integral function of the integral variable x then
- (a) λ is an integer and μ is a rational fraction
 (b) λ and μ are integers
 (c) μ is an integer and λ is a rational fraction
 (d) λ and μ are rational fractions
- Q 2. Let $f(x) = ax^2 + bx + c$, where a, b, c are rational, and $f: \mathbb{Z} \rightarrow \mathbb{Z}$ where $\mathbb{Z}$ is the set of integers. Then $a + b$ is
- (a) a negative integer (b) an integer
 (c) nonintegral rational number (d) none of these
- Q 3. If $f(x) = \cos [\pi]x + \cos [\pi x]$, where $[y]$ is the greatest integer function of y then $f\left(\frac{\pi}{2}\right)$ is equal to
- (a) $\cos 3$ (b) 0 (c) $\cos 4$ (d) none of these
- Q 4. Let $f(x) = \sin (\tan^{-1}x)$. Then $[f(-\sqrt{3})]$, where $[.]$ denotes the greatest integer function, is
- (a) $-\frac{\sqrt{3}}{2}$ (b) 0 (c) -1 (d) none of these
- Q 5. If $f(x) = \frac{x-1}{x+1}$ then $f(ax)$ in term of $f(x)$ is equal to
- (a) $\frac{f(x)+a}{1+af(x)}$ (b) $\frac{(a-1)f(x)+a+1}{(a+1)f(x)+a-1}$ (c) $\frac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}$ (d) none of these
- Q 6. Let $f(1) = 1$ and $f(n) = 2 \sum_{r=1}^{n-1} f(r)$. Then $\sum_{n=1}^m f(n)$ is equal to
- (a) $3^m - 1$ (b) 3^m (c) $3^m - 1$ (d) none of these
- Q 7. If $f(x+1) + f(x-1) = 2f(x)$ and $f(0) = 0$ then $f(n)$, $n \in \mathbb{N}$, is
- (a) $nf(1)$ (b) $\{f(1)\}^n$ (c) 0 (d) none of these
- Q 8. If $f(x+1) + bf\left(\frac{1}{x+1}\right) = x$, $x \neq -1$, $a \neq b$ then $f(2)$ is equal to
- (a) $\frac{2a+b}{2(a^2-b^2)}$ (b) $\frac{a}{a^2-b^2}$ (c) $\frac{a+2b}{a^2-b^2}$ (d) none of these

Q 9. Let f be a function satisfying $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. If $f(1) = k$ then $f(n)$, $n \in \mathbb{N}$, is equal to

- (a) k^n (b) nk (c) n^k (d) none of these

Q 10. Let f be a function satisfying $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$. If $f(1) = 3$ then $\sum_{r=1}^n f(r)$ is equal to

- (a) $\frac{3}{2}(3^n - 1)$ (b) $\frac{3}{2}n(n+1)$ (c) $3^{n+1} - 3$ (d) none of these

Q 11. If $f(x + y) = f(x) + f(y) - xy - 1$ for all x, y , and $f(1) = 1$ then the number of solutions of $f(n) = n$, $n \in \mathbb{N}$, is

- (a) one (b) two (c) four (d) none of these

Q 12. Let $f(x) = 1 + [x]$, $x < -1$

$[x]$, $x \geq -1$, where $[.]$ denotes the greatest integer function.

Then $f\{f(-2, 3)\}$ is equal to

- (a) 4 (b) 2 (c) -3 (d) 3

Q 13. The domain of the function

$$y = \log_{10} \log_{10} \log_{10} \dots \log_{10} x \text{ is}$$

- (a) $[10^n, +\infty)$ (b) $(10^{n-1}, +\infty)$ (c) $(10^{n-2}, +\infty)$ (d) none of these

Q 14. The largest set of real values of x for which

$$f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}}$$

is real function is

- (a) $[1, 2) \cup (2, 5]$ (b) $(2, 5]$ (c) $[3, 4]$ (d) none of these

Q 15. Let $f(x) = (x^{12} - x^9 + x^4 - x + 1)^{-1/2}$. The domain of the function is

- (a) $(1, +\infty)$ (b) $(-\infty, -1)$ (c) $(-1, 1)$ (d) $(-\infty, +\infty)$

Q 16. The domain of the function $f(x) = \sqrt{x - \sqrt{1 - x^2}}$ is

- (a) $\left[-1, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$ (b) $[-1, 1]$
(c) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{\sqrt{2}}, +\infty\right)$ (d) $\left[\frac{1}{\sqrt{2}}, 1\right]$

- Q 17. The domain of the function $f(x) = \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}}$ is
- (a) $\{x \mid x < 1\}$ (b) $\{x \mid x > -1\}$ (c) $[0, 1]$ (d) $[-1, 1]$
- Q 18. The domain of the function $f(x) = \log_{10} \log_{10} (1 + x^3)$ is
- (a) $(-1, +\infty)$ (b) $(0, +\infty)$ (c) $[0, +\infty)$ (d) $(-1, 0)$
- Q 19. The domain of the function $f(x) = \sqrt{x^2 - [x]^2}$, where $[x]$ = the greatest integer less than or equal to x , is
- (a) $\mathbb{R}$ (b) $[0, +\infty)$ (c) $(-\infty, 0]$ (d) none of these
- Q 20. The domain of $f(x) = \frac{1}{\sqrt{|\cos x| + \cos x}}$ is
- (a) $[-2n\pi, 2n\pi]$ (b) $(2n\pi, 2n+1\pi)$ (c) $\left(\frac{(4n+1)\pi}{2}, \frac{(4n+3)\pi}{2}\right)$ (d) $\left(\frac{(4n-1)\pi}{2}, \frac{(4n+1)\pi}{2}\right)$
- Q 21. The domain of $f(x) = \sqrt{\log_{x^2-1}(x)}$ is
- (a) $(\sqrt{2}, +\infty)$ (b) $(0, +\infty)$ (c) $(1, +\infty)$ (d) none of these
- Q 22. The domain of the function $f(x) = {}^{16-x}C_{2x-1} + {}^{20-3x}P_{4x-5}$, where the symbols have their usual meanings, is the set
- (a) $\{1, 2, 3, 4, 5\}$ (b) $\{2, 3, 4\}$ (c) $\{2, 3\}$ (d) none of these
- Q 23. The domain of $f(x) = \sin^{-1}\left(\frac{1+x^2}{2x}\right) + \sqrt{1-x^2}$ is
- (a) $\{1\}$ (b) $(-1, 1)$ (c) $\{1, -1\}$ (d) none of these
- Q 24. The domain of the function $f(x) = \sqrt{\sec^{-1}\left\{\frac{1-|x|}{2}\right\}}$ is
- (a) $(-\infty, -3] \cup [3, +\infty)$ (b) $[3, +\infty)$ (c) $\emptyset$ (d) $\mathbb{R}$
- Q 25. The function $f(x) = \sqrt{e^{\cos^{-1}(\log_4 x^2)}}$ is real valued. It is defined if
- (a) $x \in \left[\frac{1}{2}, 2\right]$ (b) $x \in \left[-2, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 2\right]$ (c) $x \in \left[-2, -\frac{1}{2}\right]$ (d) none of these
- Q 26. The domain of the real-valued function $f(x) = \log_e |\log_e x|$ is

- (a) $(1, +\infty)$ (b) $(0, +\infty)$ (c) $(e, +\infty)$ (d) none of these

Q 27. If $[.]$ denotes the greatest integer function then the domain of the real valued function $\log_{[x+1/2]} |x^2 - x - 2|$ is

- (a) $\left[\frac{3}{2}, +\infty\right)$ (b) $\left[\frac{3}{2}, 2\right) \cup (2, +\infty)$ (c) $(e, +\infty)$ (d) none of these

Q 28. The domain of the function $f(x) = \log_e (x - [x])$, where $[.]$ denotes the greatest integer function, is

- (a) $\mathbb{R}$ (b) $\mathbb{R} - \mathbb{Z}$ (c) $(0, +\infty)$ (d) none of these

Q 29. The domain of the function $f(x) = \sin^{-1} (x + [x])$, where $[.]$ denote the greatest integer function, is

- (a) $[0, 1)$ (b) $[-1, 1]$ (c) $(-1, 0)$ (d) none of these

Q 30. Let $f(x) = \log_{x^2} 25$ and $g(x) = \log_x 5$ then $f(x) = g(x)$ holds for x belonging to

- (a) $\mathbb{R}$ (b) $(0, 1) \cup (1, +\infty)$ (c) ϕ (d) none of these

Q 31. Let $f(x) = \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}$ and $g(x) = \sec^2 x - \tan^2 x$. The two functions are equal over the set

- (a) ϕ (b) $\mathbb{R}$ (c) $\mathbb{R} = \left\{ x \mid x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \right\}$ (d) none of these

Q 32. The range of the function $f(x) = x^2 + \frac{1}{x^2 + 1}$ is

- (a) $[1, +\infty)$ (b) $[2, +\infty)$ (c) $\left[\frac{3}{2}, +\infty\right)$ (d) none of these

Q 33. Let $f(x) = \cos^{-1} \left(\frac{x^2}{1+x^2} \right)$. The range of f is

- (a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (c) $\left[-\frac{\pi}{2}, 0\right]$ (d) none of these

Q 34. The range of the real-valued function $f(x) = \sqrt{9 - x^2}$ is

- (a) $[0, 3]$ (b) $[-3, 3]$ (c) $[-3, 0]$ (d) none of these

Q 35. The range of the function $f(x) = |x - 1| + |x - 2|$, $-1 \leq x \leq 3$, is

- (a) $[1, 3]$ (b) $[1, 5]$ (c) $[3, 5]$ (d) none of these

Q 36. The range of the function $y = \log_3 (5 + 4x - x^2)$ is

- (a) $(0, 2]$ (b) $(-\infty, 2]$ (c) $(0, 9]$ (d) none of these

Q 37. Let $f: \{x, y, z\} \rightarrow \{a, b, c\}$ be a one-one function and only one of the conditions (i) $f(x) \neq b$, (ii) $f(y) = b$, (iii) $f(z) \neq a$ is true then the function f is given by the set

- (a) $\{(x, a), (y, b), (z, c)\}$ (b) $\{(x, a), (y, c), (z, b)\}$ (c) $\{(x, b), (y, a), (z, c)\}$ (d) $\{(x, c), (y, b), (z, a)\}$

Q 38. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 - 6x^2 + 11x - 6$. Then

- (a) f is one-one and into (b) f is many-one and into
(c) f is one-one and onto (d) f is many-one and onto

Q 39. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 + x^2 + 3x + \sin x$. Then

- (a) f is one-one and into (b) f is one-one and onto
(c) f is many-one and into (d) f is many-one and onto

Q 40. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 6^x + 6^{|x|}$ is

- (a) one-one and onto (b) many-one and onto
(c) one-one and into (d) many-one and into

Q 41. If the real-valued function $f(x) = px + \sin x$ is a bijective function then the set of possible value of $p \in \mathbb{R}$ is

- (a) $\mathbb{R} - \{0\}$ (b) $\mathbb{R}$ (c) $(0, +\infty)$ (d) none of these

Q 42. Let $f(x) = 2x + |\cos x|$. Then f is

- (a) one-one and into (b) one-one and onto (c) many-one and into (d) many-one and onto

Q 43. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ given by $f(x) = \frac{x^2 - 4}{x^2 + 1}$. Then $f(x)$ is

- (a) one-one and into (b) one-one and onto (c) many-one and into (d) many-one and onto

Q 44. Let $f: \mathbb{R} \rightarrow A = \left\{ y \mid 0 \leq y < \frac{\pi}{2} \right\}$ be a function such that $f(x) = \tan^{-1}(x^2 + x + k)$, where k is a constant.

The value of k for which f is an onto function, is

- (a) 1 (b) 0 (c) $\frac{1}{4}$ (d) none of these

Q 45. $f(x) = x + \sqrt{x^2}$ is a function from $\mathbb{R} \rightarrow \mathbb{R}$. Then $f(x)$ is

- (a) injective (b) surjective (c) bijective (d) none of these

Q 46. Which of the following is an even function ?

Here $[.]$ denotes the greatest integer function and f is any function.

- (a) $[x] - x$ (b) $f(x) - f(-x)$ (c) $e^{3-2x} \cdot \tan^2 x$ (d) $f(x) + f(-x)$

Q 47. Let $f(x) = |x - 2| + |x - 3| + |x - 4|$ and $g(x) = f(x + 1)$. Then

- (a) $g(x)$ is an even function (b) $g(x)$ is an odd function
(c) $g(x)$ is neither even nor odd (d) $g(x)$ is periodic

Q 48. $f(x) = \log_{10}(x + \sqrt{x^2 + 1})$ is

- (a) an odd function (b) a periodic function (c) an even function (d) none of these

Q 49. A function whose graph is symmetrical about the y-axis is given by

- (a) $f(x) = \log_e(x + \sqrt{x^2 + 1})$ (b) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$
(c) $f(x) = \cos x + \sin x$ (d) none of these

Q 50. A function whose graph is symmetrical about the origin is given by

- (a) $f(x) = e^x + e^{-x}$ (b) $f(x) = \log_e x$ (c) $f(x + y) = f(x) + f(y)$ (d) none of these

Q 51. Let $f(x) = 4, x < -1$

$$-4x, -1 \leq x \leq 0.$$

If $f(x)$ is an even function in $\mathbb{R}$ then the definition of $f(x)$ in $(0, +\infty)$ is

- (a) $f(x) = 4x, 0 < x \leq 1$ (b) $f(x) = 4x, 0 < x \leq 1$ (c) $f(x) = 4, 0 < x \leq 1$ (d) none of these
 $4, x > 1$ $-4, x > 1$ $4x, x > 1$

Q 52. If $f(x) = x^2 \sin \frac{\pi x}{2}, |x| < 1$

$x |x|, |x| \geq 1$ then $f(x)$ is

- (a) an even function (b) an odd function (c) a periodic function (d) none of these

Q 53. The period of the function $f(x) = \left| \sin \frac{x}{2} \right| + |\cos x|$ is

- (a) 2π (b) π (c) 4π (d) none of these

- Q 54. If $f(x)$ is a periodic function of the period k then $f(kx + a)$, where a is a constant, is a periodic function of the period
- (a) k (b) 1 (c) $\frac{k}{a}$ (d) none of these
- Q 55. The period of the function $f(x) = 4\cos(2x + 3)$ is
- (a) 2π (b) $\frac{\pi}{2}$ (c) π (d) none of these
- Q 56. The period of the function $f(x) = 3\sin\frac{\pi x}{3} + 4\cos\frac{\pi x}{4}$ is
- (a) 6 (b) 24 (c) 8 (d) 2π
- Q 57. Let $f(x) = \cos\sqrt{p}x$, where $p = [a]$ = the greatest integer less than or equal to a . If the period of $f(x)$ is π then
- (a) $a \in [4, 5]$ (b) $a = 4, 5$ (c) $a \in [4, 5)$ (d) none of these
- Q 58. Let $f(x) = \cos 3x + \sin\sqrt{3}x$. Then $f(x)$ is
- (a) a periodic function of period 2π (b) a periodic function of period $\sqrt{3}\pi$
(c) not a periodic function (d) none of these
- Q 59. The function $f(x) = \sin\frac{\pi x}{n!} - \cos\frac{\pi x}{(n+1)!}$ is
- (a) not periodic (b) periodic, with period $2(n!)$
(c) periodic, with period $(n+1)$ (d) none of these
- Q 60. The function $f(x) = x - [x] + \cos x$, where $[x]$ = the greatest integer less than or equal to x , is a
- (a) periodic function of indeterminate period (b) periodic function of period 2π
(c) nonperiodic function (d) periodic function of period 1
- Q 61. Let $f(x) = nx + n - [nx + n] + \tan\frac{\pi x}{2}$, where $[x]$ is the greatest integer $\leq x$ and $n \in \mathbb{N}$. It is
- (a) a periodic function of period 1 (b) a periodic function of period 4
(c) not periodic (d) a periodic function of period 2

Q 62. Let $f(x) = x(2 - x)$, $0 \leq x \leq 2$. If the definition of f is extended over the set $\mathbb{R} - [0, 2]$ by $f(x + 2) = f(x)$ the f is a

- (a) periodic function of period 1 (b) nonperiodic function
(c) periodic function of period 2 (d) periodic function of period $\frac{1}{2}$

Q 63. If $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cdot \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$ then $(g \circ f)(x)$ is

- (a) a polynomial of the first degree in $\sin x$, $\cos x$
(b) a constant function
(c) a polynomial of the second degree in $\sin x$, $\cos x$
(d) none of these

Q 64. If $f(x) = x^n$, $n \in \mathbb{N}$ and $(g \circ f)(x) = ng(x)$ then $g(x)$ can be

- (a) $n|x|$ (b) $3 \cdot \sqrt[3]{x}$ (c) e^x (d) $\log|x|$

Q 65. If $g\{f(x)\} = |\sin x|$ and $f\{g(x)\} = (\sin \sqrt{x})^2$ then

- (a) $f(x) = \sin^2 x, g(x) = \sqrt{x}$ (b) $f(x) = \sin x, g(x) = |x|$
(c) $f(x) = x^2, g(x) = \sin \sqrt{x}$ (d) f and g cannot be determined

Q 66. If $f(x) = \frac{1}{1-x}$, $x \neq 0, 1$, then the graph of the function $y = f\{f(f(x))\}$, $x > 1$, is

- (a) a circle (b) an ellipse (c) a straight line (d) a pair of straight lines

Q 67. If $f(x)$ is a polynomial function of the second degree such that $f(-3) = 6$, $f(0) = 6$ and $f(2) = 11$ then the graph of the function $f(x)$ cuts the ordinate $x = 1$ at the point

- (a) (1, 8) (b) (1, 4) (c) (1, -2) (d) none of these

Q 68. Let $f(x)$ be a function whose domain is $[-5, 7]$. Let $g(x) = |2x + 5|$. Then the domain of $(f \circ g)(x)$ is

- (a) $[-5, 1]$ (b) $[-4, 0]$ (c) $[-6, 1]$ (d) none of these

Q 69. Let $f : (-\infty, 1] \rightarrow (-\infty, 1]$ such that $f(x) = x(2 - x)$. Then $f^{-1}(x)$ is

- (a) $1 + \sqrt{1-x}$ (b) $1 - \sqrt{1-x}$ (c) $\sqrt{1-x}$ (d) none of these

Q 70. If $f(x) = 3x - 5$ then $f^{-1}(x)$

(a) is given by $\frac{1}{3x-5}$

(b) is given by $\frac{x+5}{3}$

(c) does not exist because f is not one-one

(d) does not exist because f is not onto

Q 71. If the function $f: [1, +\infty) \rightarrow [1, +\infty)$ is defined by $f(x) = 2^{x(x-1)}$ then $f^{-1}(x)$ is

(a) $\left(\frac{1}{2}\right)^{x(x-1)}$

(b) $\frac{1}{2}(1 + \sqrt{1 + 4\log_2 x})$

(c) $\frac{1}{2}(1 - \sqrt{1 + 4\log_2 x})$

(d) not defined

Q 72. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x) = x - [x]$, where $[y]$ denotes the greatest integer less than or equal to y , then $f^{-1}(x)$ is

(a) $\frac{1}{x - [x]}$

(b) $x - [x]$

(c) not defined

(d) none of these

Q 73. The inverse function of the function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is

(a) $\frac{1}{2} \log \frac{1+x}{1-x}$

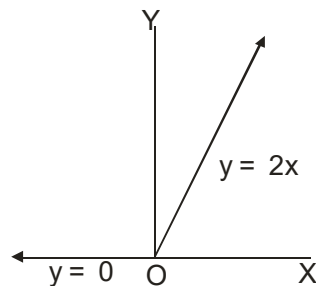
(b) $\frac{1}{2} \log \frac{2+x}{2-x}$

(c) $\frac{1}{2} \log \frac{1-x}{1+x}$

(d) none of these

Q 74. The graph of a real-valued function $f(x)$ is the following.

The function is



(a) $f(x) = x - |x|$

(b) $f(x) = x + |x|$

(c) $f(x) = 2x$

(d) none of these

Q 75. If $f(x+y, x-y) = xy$ then the arithmetic mean of $f(x, y)$ and $f(x, y)$ is

(a) x

(b) y

(c) 0

(d) none of these

Q 76. The graph of the function $y = f(x)$ is symmetrical about the line $x = 2$.

Then

(a) $f(x+2) = f(x-2)$

(b) $f(2+x) = f(2-x)$

(c) $f(x) = f(-x)$

(d) none of these

Choose the correct options. One or more options may be correct.

Q 77. Let $f(x) = x^2$, $0 < x < 2$

$$2x - 3, 2 \leq x < 3$$

$$x + 2, x \geq 3. \text{ Then}$$

(a) $f\left\{f\left(f\left(\frac{3}{2}\right)\right)\right\} = f\left(\frac{3}{2}\right)$

(b) $1 + f\left\{f\left(f\left(\frac{5}{2}\right)\right)\right\} = f\left(\frac{5}{2}\right)$

(c) $f\{f(1)\} = f(1) = 1$

(d) none of these

Q 78. If $f(x) = \cos^2 x + \cos^2\left(\frac{\pi}{3} + x\right) - \cos x \cdot \cos\left(\frac{\pi}{3} + x\right)$ then

(a) $f(x)$ is an even function

(b) $f\left(\frac{\pi}{8}\right) = f\left(\frac{\pi}{4}\right)$

(c) $f(x)$ is a constant function

(d) $f(x)$ is not periodic function

Q 79. If one of the roots of $x^2 + f(a) \cdot x + a = 0$ is equal to the third power of the other for real a then

(a) the domain of the real-valued function f is the set of non-negative real numbers

(b) $f(x) = -x^{1/4}(1 + x^{1/2})$

(c) $f(x) = x^{1/4} + x^{3/4}$

(d) none of these

Q 80. If f is an even function defined on the interval $(-5, 5)$ then a value of x satisfying the equation

$$f(x) = f\left(\frac{x+1}{x+2}\right) \text{ is}$$

(a) $\frac{-1+\sqrt{5}}{2}$

(b) $\frac{-2+\sqrt{5}}{2}$

(c) $\frac{-1-\sqrt{5}}{2}$

(d) $\frac{-3-\sqrt{5}}{2}$

Q 81. Let $f(x) = [x]$ = the greatest integer less than or equal to x and $g(x) = x - [x]$. Then for any two real numbers x and y

(a) $f(x+y) = f(x) + f(y)$ (b) $g(x+y) = g(x) + g(y)$ (c) $f(x+y) = f(x) + f\{y + g(x)\}$ (d) none of these

Q 82. Let $x \in \mathbb{N}$ and let x be a perfect square. Let $f(x)$ = the quotient when x is divided by 5 and $g(x)$ = the remainder when x is divided by 5. Then $\sqrt{x} = f(x) + g(x)$ holds for x equal to

(a) 0

(b) 16

(c) 25

(d) none of these

Q 83. If $f(x) = 27x^3 + \frac{1}{x^3}$ and α, β are the roots of $3x + \frac{1}{x} = 2$ then

(a) $f(\alpha) = f(\beta)$

(b) $f(\alpha) = 10$

(c) $f(\beta) = -10$

(d) none of these

Q 84. If $f(x) = \sin^{-1}(\sin x)$ then

(a) $f(x) = \pi - x, 0 \leq x \leq \frac{\pi}{2}$

(b) $f(x) = \pi - x, \frac{\pi}{2} \leq x \leq \pi$

(c) $f(x) = x, 0 \leq x \leq \pi$

(d) $f(x) = -x, -\frac{\pi}{2} \leq x \leq 0$

Q 85. If $e^x + e^{f(x)} = e$ then for $f(x)$

(a) domain = $(-\infty, 1)$

(b) range = $(-\infty, 1)$

(c) domain = $(-\infty, 0]$

(d) range = $(-\infty, 1]$

Q 86. If $f(x)$ is an odd function then

(a) $\frac{f(-x) + f(x)}{2}$ is an even function

(b) $[|f(x)| + 1]$ is even, where $[x]$ = the greatest integer $\leq x$

(c) $\frac{f(x) - f(-x)}{2}$ is neither even nor odd

(d) none of these

Q 87. Let $f(x) = \sec^{-1}[1 + \cos^2 x]$ where $[.]$ denotes the greatest integer function. Then

(a) the domain of f is $\mathbb{R}$

(b) the domain of f is $[1, 2]$

(c) the range of f is $[1, 2]$

(d) the range of f is $\{\sec^{-1} 1, \sec^{-1} 2\}$

Q 88. If $f(x)$ and $g(x)$ are two functions of x such that $f(x) + g(x) = e^x$ and $g(x) - f(x) = e^{-x}$ then

(a) $f(x)$ is an odd function

(b) $g(x)$ is an odd function

(c) $f(x)$ is an even function

(d) $g(x)$ is an even function

Q 89. Let $f(x) = 4 \cos \sqrt{x^2 - \frac{\pi^2}{9}}$. Then

(a) the domain of f is $\left[\frac{\pi}{3}, +\infty\right)$

(b) the range of f is $[-1, 1]$

(c) the domain of f is $\left(-\infty, -\frac{\pi}{3}\right] \cup \left[\frac{\pi}{3}, +\infty\right)$

(d) the range of f is $[-4, 4]$

Q 90. Let $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Then

(a) $f(x)$ is an even function

(b) $f(x)$ is an odd function

(c) $f(0) = 0$

(d) $f(n) = nf(1), n \in \mathbb{N}$

Q 91. Let $f(x) = [x]^2 + [x + 1] - 3$, where $[x]$ = the greatest integer $\leq x$. Then

- (a) $f(x)$ is a many-one and into function (b) $f(x) = 0$ for infinite number of values of x
 (c) $f(x) = 0$ for only two real values (d) none of these

Q 92. Let f and g be functions from the interval $[0, \infty)$ to the interval $[0, \infty)$ f being an increasing function and g being a decreasing function. If $f\{g(0)\} = 0$ then

- (a) $f\{g(x)\} \geq f\{g(0)\}$ (b) $g\{f(x)\} \leq g\{f(0)\}$ (c) $f\{g(2)\} = 0$ (d) none of these

1b	2b	3c	4c	5c	6c	7a	8a	9b	10a
11a	12d	13d	14b	15d	16d	17d	18b	19d	20d
21a	22c	23c	24a	25b	26d	27b	28b	29a	30b
31c	32a	33d	34a	35b	36b	37c	38d	39b	40c
41a	42b	43c	44c	45d	46d	47c	48a	49d	50c
51a	52b	53a	54b	55c	56b	57c	58c	59d	60c
61d	62c	63b	64d	65a	66c	67a	68c	69b	70b
71b	72c	73a	74b	75c	76b	77abc	78abc	79ab	80abcd
81c	82bc	83ac	84b	85ab	86ab	87ad	88bc	89cd	90bcd
91ab	92bc								

Choose the most appropriate option (a, b, c or d).

- Q 1. If $y = (1 + x)(1 + x^2)(1 + x^4) \dots (1 + x^{2^n})$ then $\frac{dy}{dx}$ at $x = 0$ is
- (a) 1 (b) -1 (c) 0 (d) none of these
- Q 2. If $y = |\cos x| + |\sin x|$ then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is
- (a) $\frac{1-\sqrt{3}}{2}$ (b) 0 (c) $\frac{1}{2}(\sqrt{3}-1)$ (d) none of these
- Q 3. The differential coefficient of $f(\log_e x)$ with respect to x , where $f(x) = \log_e x$, is
- (a) $\frac{x}{\log_e x}$ (b) $\frac{1}{x} \log_e x$ (c) $\frac{1}{x \log_e x}$ (d) none of these
- Q 4. If $f(x) = \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x$ then $f'\left(\frac{\pi}{4}\right)$ is
- (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) 1 (d) none of these
- Q 5. If $y = \cos^{-1}(\cos x)$ then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ is equal to
- (a) 1 (b) -1 (c) $\frac{1}{\sqrt{2}}$ (d) none of these
- Q 6. If $y = \sec(\tan^{-1}x)$ then $\frac{dy}{dx}$ at $x = 1$ is equal to
- (a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$ (c) 1 (d) none of these
- Q 7. If $x^y = e^{x-y}$ then $\frac{dy}{dx}$ at $x = 1$ is equal to
- (a) 0 (b) -2 (c) 1 (d) none of these
- Q 8. If $xe^{xy} - y = \sin^2 x$ then $\frac{dy}{dx}$ at $x = 0$ is
- (a) 0 (b) 1 (c) -1 (d) none of these
- Q 9. If $x = e^{y+e^{y+\dots\infty}}$ then $\frac{dy}{dx}$ is

- (a) $\frac{x}{1+x}$ (b) $\frac{1}{x}$ (c) $\frac{1-x}{x}$ (d) none of these

Q 10. If $y = \log_2(\log_e x)$ then $\frac{dy}{dx}$ is

- (a) $\frac{1}{x} \log_2 e \cdot \log_x e$ (b) $\frac{1}{x} \log_2 x$ (c) $\frac{1}{x} \log_e x$ (d) none of these

Q 11. If $f'(x) = \sqrt{2x^2 - 1}$ and $y = f(x^2)$ then $\frac{dy}{dx}$ at $x = 1$ is

- (a) 2 (b) 1 (c) -2 (d) none of these

Q 12. If $y = \cos^{-1}\left(\frac{5\cos x - 12\sin x}{13}\right); x \in \left(0, \frac{\pi}{2}\right)$, then $\frac{dy}{dx}$ is equal to

- (a) 1 (b) -1 (c) 0 (d) none of these

Q 13. If $y = \tan^{-1} \sqrt{\frac{x+1}{x-1}}$ then $\frac{dy}{dx}$ is equal to

- (a) $\frac{-1}{2|x|\sqrt{x^2-1}}$ (b) $\frac{-1}{2x\sqrt{x^2-1}}$ (c) $\frac{1}{2x\sqrt{x^2-1}}$ (d) none of these

Q 14. If $x^y \cdot y^x = 16$ then $\frac{dy}{dx}$ at $(2, 2)$ is

- (a) 1 (b) -1 (c) 0 (d) none of these

Q 15. If $y = x^{1/x}$, the value of $\frac{dy}{dx}$ at $x = e$ is

- (a) 1 (b) 0 (c) -1 (d) none of these

Q 16. The derivative of $\tan^{-1} \frac{2x}{1-x^2}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$ is

- (a) $\frac{1}{1+x^2}$ (b) $\frac{1}{1-x^2}$ (c) 0 (d) 1

Q 17. Let the function $y = f(x)$ be given by $x = t^5 - 5t^3 - 20t + 7$ and $y = 4t^3 - 3t^2 - 18t + 3$, where $t \in (-2, 2)$. Then $f'(x)$ at $t = 1$ is

- (a) $\frac{5}{2}$ (b) $\frac{2}{5}$ (c) $\frac{7}{5}$ (d) none of these

Q 18. If $f(x) = \cos\left\{\frac{\pi}{2}[x] - x^3\right\}$, $1 < x < 2$, and $[x]$ = the greatest integer $\leq x$, then $f'\left(\sqrt[3]{\frac{\pi}{2}}\right)$ is equal to

- (a) 0 (b) $3\left(\frac{\pi}{2}\right)^{2/3}$ (c) $-3\left(\frac{\pi}{2}\right)^{3/2}$ (d) none of these

Q 19. If $y = \sin x^\circ$ and $x = \cos x$ then $\frac{dy}{dz}$ is equal to

- (a) $-\operatorname{cosec} x \cdot \cos x$ (b) $\frac{\pi}{180} \operatorname{cosec} \frac{\pi x}{180} \cdot \cos x$
(c) $-\frac{\pi}{180} \operatorname{cosec} \cdot \cos \frac{\pi x}{180}$ (d) none of these

Q 20. The derivative of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\tan^{-1} x$ is

- (a) $\frac{\sqrt{1+x^2}-1}{x^2}$ (b) 1 (c) $\frac{1}{1+x^2}$ (d) none of these

Q 21. The differential coefficient of $\operatorname{cosec}^{-1} \frac{1}{2x^2-1}$ with respect to $\sqrt{1-x^2}$ at $x = \frac{1}{2}$ is

- (a) -4 (b) 4 (c) -1 (d) none of these

Q 22. If the prime sign (') represents differentiation w.r.t. x and $f'(x) = \sin x + \sin 4x \cdot \cos x$ then

$f'\left(2x^2 + \frac{\pi}{2}\right)$ at $x = \sqrt{\frac{\pi}{2}}$ is equal to

- (a) 0 (b) -1 (c) $-2\sqrt{2\pi}$ (d) none of these

Q 23. If $u = f(x^3)$, $v = g(x^2)$, $f'(x) = \cos x$ and $g'(x) = \sin x$ then $\frac{du}{dv}$ is

- (a) $\frac{3}{2} x \cdot \cos x^3 \cdot \operatorname{cosec} x^2$ (b) $\frac{2}{3} \sin x^3 \cdot \sec x^2$ (c) $\tan x$ (d) none of these

Q 24. If $y = f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x^2$ then $\frac{dy}{dx}$ is

- (a) $\sin\left(\frac{2x-1}{x^2+1}\right)^2$ (b) $\frac{2(1+x-x^2)}{(x^2+1)^2} \sin\left(\frac{2x-1}{x^2+1}\right)^2$
(c) $\frac{2(2x-1)}{x^2+1} \sin\left(\frac{2x-1}{x^2+1}\right)^2$ (d) none of these

- Q 25. If $e^x = \frac{\sqrt{1+t} - \sqrt{1-t}}{\sqrt{1+t} + \sqrt{1-t}}$ and $\tan \frac{y}{2} = \sqrt{\frac{1-t}{1+t}}$ then $\frac{dy}{dx}$ at $t = \frac{1}{2}$ is
- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 0 (d) none of these
- Q 26. If $t(1+x^2) = x$ and $x^2 + t^2 = y$ then $\frac{dy}{dx}$ at $x = 2$ is
- (a) $\frac{88}{125}$ (b) $\frac{488}{125}$ (c) 1 (d) none of these
- Q 27. Let $f(x)$ be a polynomial function of the second degree. If $f(1) = f(-1)$ and a_1, a_2, a_3 are in AP then $f'(a_1), f'(a_2), f'(a_3)$ are in
- (a) AP (b) GP (c) HP (d) none of these
- Q 28. If $P(x)$ is a polynomial such that $P(x^2 + 1) = \{P(x)\}^2 + 1$ and $P(0) = 0$ then $P'(0)$ is equal to
- (a) 1 (b) 0 (c) -1 (d) none of these
- Q 29. If $5f(x) + 3f\left(\frac{1}{x}\right) = x + 2$ and $y = xf(x)$ then $\left(\frac{dy}{dx}\right)_{x=1}$ is equal to
- (a) 14 (b) $\frac{7}{8}$ (c) 1 (d) none of these
- Q 30. If for all x, y the function f is defined by
- $$f(x) + f(y) + f(x) \cdot f(y) = 1 \text{ and } f(x) > 0 \text{ then}$$
- (a) $f'(x)$ does not exist (b) $f'(x) = 0$ for all x (c) $f'(0) < f'(1)$ (d) none of these
- Q 31. There exists a function $f(x)$ satisfying $f(0) = 1, f'(0) = -1, f(x) > 0$ for all x and
- (a) $f'(x) < 0$ for all x (b) $-1 < f''(x) < 0$ for all x (c) $-2 \leq f''(x) \leq -1$ for all x (d) $f''(x) \leq -2$ for all x
- Q 32. If g is the inverse function of f and $f'(x) = \sin x$ then $g'(x)$ is
- (a) $\operatorname{cosec}\{g(x)\}$ (b) $\sin\{g(x)\}$ (c) $-\frac{1}{\sin\{g(x)\}}$ (d) none of these
- Q 33. Let $f(x)$ be a polynomial function of degree 2 and $f(x) > 0$ for all $x \in \mathbb{R}$. If $g(x) = f(x) + f'(x) + f''(x)$ then for any x
- (a) $g(x) < 0$ (b) $g(x) > 0$ (c) $g(x) = 0$ (d) $g(x) \geq 0$

Q 34. If $y^2 = P(x)$ is a polynomial of degree 3 then $2 \frac{d}{dx} \left(y^3 \frac{d^2 y}{dx^2} \right)$ equals

- (a) $P'''(x) + P'(x)$ (b) $P''(x) \cdot P'''(x)$ (c) $P(x) \cdot P'''(x)$ (d) none of these

Q 35. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$,

where p is a constant. Then $\frac{d^3}{dx^3} \{f(x)\}$ at $x = 0$ is

- (a) p (b) $p + p^2$ (c) $p + p^3$ (d) independent of p

Q 36. If $y = \sin 2x$ then $\frac{d^6 y}{dx^6}$ at $x = \frac{\pi}{2}$ is equal to

- (a) -64 (b) 0 (c) 64 (d) none of these

Q 37. $x = t \cos t$, $y = t + \sin t$ then $\frac{d^2 x}{dy^2}$ at $t = \frac{\pi}{2}$ is equal to

- (a) $\frac{\pi + 4}{2}$ (b) $-\frac{\pi + 4}{2}$ (c) -2 (d) none of these

Q 38. If $y = at^2$, $x = 2at$, where a is a constant, then $\frac{d^2 y}{dx^2}$ at $x = \frac{1}{2}$ is

- (a) $\frac{1}{2a}$ (b) 1 (c) $2a$ (d) none of these

Choose the correct options. One or more options may be correct.

Q 39. Let $f(x) = (ax + b)\cos x + (cx + d)\sin x$ and $f'(x) = x \cos x$ be an identity in x . Then

- (a) $a = 5$ (b) $b = 1$ (c) $c = 1$ (d) $d = -5$

Q 40. Let $f(x) = 2\tan^{-1}x + \sin^{-1} \frac{2x}{1+x^2}$. Then

- (a) $f'(2) = f'(3)$ (b) $f'(2) = 0$ (c) $f'\left(\frac{1}{2}\right) = \frac{16}{5}$ (d) $f'\left(\frac{1}{2}\right) = 0$

Q 41. If $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for all $x \in \mathbb{R}$ then

- (a) $f(0) + f(2) = f(1)$ (b) $f(0) + f(3) = 0$ (c) $f(1) + f(3) = f(2)$ (d) none of these

Q 42. If $f(x - y)$, $f(x) \cdot f(y)$ and $f(x + y)$ are in AP for all x, y , and $f(0) \neq 0$, then

- (a) $f(2) = f(-2)$ (b) $f(3) + f(-3) = 0$ (c) $f'(2) + f'(-2) = 0$ (d) $f'(3) = f'(-3)$

Q 43. Let $f(-x) = f(x)$. Then $f'(x)$ must be

- (a) an even function (b) an odd function (c) a periodic function (d) neither even nor off

Q 44. If $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in \mathbb{N}$ and $f_0(x) = x$ then $\frac{d}{dx}\{f_n(x)\}$ is equal to

- (a) $f_n(x) \cdot \frac{d}{dx}\{f_{n-1}(x)\}$ (b) $f_n(x) \cdot f_{n-1}(x)$ (c) $f_n(x) \cdot f_{n-1}(x) \cdot \dots \cdot f_2(x) \cdot f_1(x)$ (d) none of these

Q 45. Let $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1) \cdot x^2 + xf'(x) + f''(x)$ then

- (a) $f'(1) + f'(2) = 0$ (b) $g'(2) = g'(1)$ (c) $g''(2) + f''(3) = 0$ (d) none of these

Answers

1a	2c	3c	4a	5b	6a	7a	8b	9c	10a
11a	12a	13a	14b	15b	16d	17b	18a	19c	20d
21a	22c	23a	24b	25a	26b	27a	28a	29b	30b
31a	32a	33b	34c	35d	36b	37b	38a	39bc	40abc
41abc	42ac	43b	44ac	45abc					

Limits, Indeterminate Forms

Choose the most appropriate option (a, b, c or d).

- Q 1. $\lim_{x \rightarrow 1} \frac{(\sqrt{x}-1)(2x-3)}{2x^2+x-3}$ is equal to
(a) $\frac{1}{10}$ (b) $-\frac{1}{10}$ (c) 1 (d) none of these
- Q 2. $\lim_{x \rightarrow 2} \frac{\sqrt{x-2} + \sqrt{x} - \sqrt{2}}{\sqrt{x^2-4}}$ is equal to
(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) none of these
- Q 3. $\lim_{x \rightarrow \infty} (\sqrt{x+\sqrt{x}} - \sqrt{x})$ is equal to
(a) 1 (b) 0 (c) $\frac{1}{2}$ (d) none of these
- Q 4. $\lim_{n \rightarrow \infty} \frac{a^n + b^n}{a^n - b^n}$, where $a > b > 1$, is equal to
(a) -1 (b) 1 (c) 0 (d) none of these
- Q 5. $\lim_{x \rightarrow 0} \frac{3^x - 1}{\sqrt{x+1} - 1}$ is equal to
(a) $\log_e 9$ (b) $\log_e 3$ (c) 0 (d) 1
- Q 6. $\lim_{n \rightarrow \infty} \frac{4^{1/n} - 1}{3^{1/n} - 1}$ is equal to
(a) $\log_4 3$ (b) 1 (c) $\log_3 4$ (d) none of these
- Q 7. $\lim_{x \rightarrow \infty} \frac{\cos x + \sin x}{x^2}$ is equal to
(a) 1 (b) 0 (c) ∞ (d) none of these
- Q 8. $\lim_{x \rightarrow 0} \frac{3^x - 2^x}{4^x - 3^x}$ is equal to
(a) 1 (b) -1 (c) 0 (d) none of these

Q 9. $\lim_{h \rightarrow 0} \left\{ \frac{1}{h \sqrt[3]{8+h}} - \frac{1}{2h} \right\}$ is equal to

- (a) $\frac{1}{12}$ (b) $-\frac{4}{3}$ (c) $-\frac{16}{3}$ (d) $-\frac{1}{48}$

Q 10. $\lim_{n \rightarrow \infty} \frac{n^p \sin^2(n!)}{n+1}$, $0 < p < 1$, is equal to

- (a) 0 (b) ∞ (c) 1 (d) none of these

Q 11. $\lim_{x \rightarrow 0} \frac{x \sqrt{y^2 - (y-x)^2}}{(\sqrt{8xy} - 4x^2 + \sqrt{8xy})^3}$ is equal to

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{2\sqrt{2}}$ (d) none of these

Q 12. $\lim_{x \rightarrow 0} \frac{(1+x+x^2) - e^x}{x^2}$ is equal to

- (a) 1 (b) 0 (c) $\frac{1}{2}$ (d) none of these

Q 13. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is equal to

- (a) 2 (b) -2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

Q 14. $\lim_{x \rightarrow 0} \left\{ \frac{\log_e(1+x)}{x^2} + \frac{x-1}{x} \right\}$ is equal to

- (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 1 (d) none of these

Q 15. $\lim_{n \rightarrow \infty} (0.2)^{\log_{\sqrt{6}}(1/4 + 1/8 + 1/16 + \dots \text{to } n \text{ terms})}$ is equal to $\lim_{n \rightarrow \infty} (0.2)^{\log_{\sqrt{6}}(1/4 + 1/8 + 1/16 + \dots \text{to } n \text{ terms})}$ is equal to

- (a) 2 (b) 4 (c) 8 (d) 0

Q 16. Let $s_n = \frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ to n terms. Then $\lim_{n \rightarrow \infty} s_n$ is equal to

- (a) $\frac{1}{3}$ (b) 3 (c) $\frac{1}{4}$ (d) ∞

Q 17. Let the r th term, t_r , of a series is given by $t_r = \frac{r}{1+r^2+r^4}$. Then $\lim_{n \rightarrow \infty} \sum_{r=1}^n t_r$ is

- (a) $\frac{1}{4}$ (b) 1 (c) $\frac{1}{2}$ (d) none of these

Q 18. Let $a = \min \{x^2 + 2x + 3, x \in \mathbb{R}\}$ and $b = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$. The value of $\sum_{r=0}^n a^r \cdot b^{n-r}$ is

- (a) $\frac{2^{n+1} - 1}{3 \cdot 2^n}$ (b) $\frac{2^{n+1} + 1}{3 \cdot 2^n}$ (c) $\frac{4^{n+1} - 1}{3 \cdot 2^n}$ (d) none of these

Q 19. $\lim_{x \rightarrow 1} \frac{\sum_{r=1}^n x^r - n}{x - 1}$ is equal to

- (a) $\frac{n}{2}$ (b) $\frac{n(n+1)}{2}$ (c) 1 (d) none of these

Q 20. $\lim_{x \rightarrow \infty} \frac{x - \tan x}{x \tan x}$ is equal to

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) none of these

Q 21. $\lim_{x \rightarrow 0} \sqrt{\frac{x - \sin x}{x + \sin^2 x}}$ is equal to

- (a) 1 (b) 0 (c) ∞ (d) none of these

Q 22. $\lim_{x \rightarrow 0} \frac{\log_e \cos x}{x^2}$ is equal to

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 0 (d) none of these

Q 23. $\lim_{x \rightarrow 0} \frac{\sin^{-1} x - \tan^{-1} x}{x^2}$ is equal to

- (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 0 (d) ∞

Q 24. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x(2^x - 1)}$ is equal to

- (a) $\frac{1}{2} \log_2 e$ (b) $\frac{1}{2} \log_e 2$ (c) 1 (d) none of these

- Q 25. If $f(4) = f$, $f'(4) = 1$ then $\lim_{x \rightarrow 4} \frac{2 - \sqrt{f(x)}}{2 - \sqrt{x}}$ is equal to
- (a) 0 (b) 1 (c) -1 (d) none of these
- Q 26. The graph of the function $y = f(x)$ has a unique tangent at the point $(a, 0)$ through which the graph passes. Then $\lim_{x \rightarrow a} \frac{\log_e \{1 + 6f(x)\}}{3f(x)}$ is
- (a) 1 (b) 0 (c) 2 (d) none of these
- Q 27. Let $f(x)$ be a twice-differentiable function and $f''(0) = 2$ then $\lim_{x \rightarrow 0} \frac{2f(x) - 3f(2x) + f(4x)}{x^2}$ is
- (a) 6 (b) 3 (c) 12 (d) none of these
- Q 28. If $f(x)$, $g(x)$ be differentiable functions and $f(1) = g(1) = 2$ then
- $\lim_{x \rightarrow 1} \frac{f(1)g(x) - f(x)g(1) - f(1) + g(1)}{g(x) - f(x)}$ is equal to
- (a) 0 (b) 1 (c) 2 (d) none of these
- Q 29. If $f(a) = 2$, $f'(a) = 1$, $g(a) = -1$, $g'(a) = 2$ then $\lim_{x \rightarrow a} \frac{g(x)f(a) - g(a)f(x)}{x - a}$ is
- (a) -5 (b) $\frac{1}{5}$ (c) 5 (d) none of these
- Q 30. $\lim_{n \rightarrow \infty} \left(1 + \sin \frac{a}{n}\right)^n$ is equal to
- (a) $e^{a/2}$ (b) e^a (c) e (d) e^{2a}
- Q 31. $\lim_{x \rightarrow 0} \left(\frac{1 + 5x}{1 + 3x^2}\right)^{1/x^2}$
- (a) e (b) $e^{1/2}$ (c) e^{-2} (d) none of these
- Q 32. $\lim_{x \rightarrow 0} \left\{ \tan \left(\frac{\pi}{4} - x \right) \right\}^{1/x}$ is equal to
- (a) 1 (b) e (c) e^2 (d) e^{-2}
- Q 33. $\lim_{x \rightarrow \pi/4} (2 - \tan x)^{\log \tan x}$ is equal to
- (a) 0 (b) 1 (c) e (d) e^{-1}

- Q 34. $\lim_{x \rightarrow \infty} \left(\frac{x-1}{x+1} \right)^{x+2}$ is equal to
 (a) e (b) e^{-1} (c) e^{-2} (d) none of these
- Q 35. If $\lim_{x \rightarrow \infty} \left(1 + \frac{\lambda}{x} + \frac{\mu}{x^2} \right)^{2x} = e^2$ then
 (a) $\lambda = 1, \mu = 2$ (b) $\lambda = 2, \mu = 1$ (c) $\lambda = 1, \mu = \text{any real constant}$ (d) $\lambda = \mu = 1$
- Q 36. $\lim_{\theta \rightarrow 0^+} \frac{\sin \sqrt{\theta}}{\sqrt{\sin \theta}}$ is equal to
 (a) 0 (b) 1 (c) -1 (d) none of these
- Q 37. Let $\{x\}$ denote the fractional part of x . Then $\lim_{x \rightarrow 0} \frac{\{x\}}{\tan\{x\}}$ is equal to
 (a) 1 (b) 0 (c) -1 (d) none of these
- Q 38. $\lim_{x \rightarrow \infty} \frac{\log_e [x]}{x}$, where $[.]$ denotes the greatest integer function, is
 (a) 0 (b) 1 (c) -1 (d) nonexistent
- Q 39. $\lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{x-1}$
 (a) exists and it is $\sqrt{2}$ (b) exists and it is $-\sqrt{2}$
 (c) does not exist because $x-1 \rightarrow 0$ (d) does not exist because $\text{LH lim} \neq \text{RH lim}$
- Q 40. $\lim_{x \rightarrow 1} \frac{x \sin\{x - [x]\}}{x-1}$, where $[.]$ denotes the greatest integer function, is
 (a) 0 (b) -1 (c) not existent (d) none of these
- Q 41. $\lim_{x \rightarrow 0} \frac{x[x]}{\sin |x|}$, where $[.]$ denotes the greatest integer function, is
 (a) 0 (b) 1 (c) not existent (d) none of these
- Q 42. $\lim_{x \rightarrow 2} \{[2-x] + [x-2] - x\}$
 (a) is 0 (b) is 3 (c) is -3 (d) does not exist
- Q 43. $\lim_{x \rightarrow -1} \{[x] + |x|\}$, where $[.]$ denotes the greatest integer function,

- (a) is 0 (b) is 1 (c) does not exist (d) none of these

Q 44. If $f(x) = \frac{\sin[x]}{[x]}, [x] \neq 0$

$0, [x] = 0$, where $[.]$ denotes the greatest integer function, then $\lim_{x \rightarrow 0} f(x)$ is equal to

- (a) 1 (b) 0 (c) -1 (d) none of these

Q 45. Let $f(x) = x^2 - 1, 0 < x < 2$

$2x + 3, 2 \leq x < 3$

The quadratic equation whose roots are $\lim_{x \rightarrow 2-0} f(x)$ and $\lim_{x \rightarrow 2+0} f(x)$ is

- (a) $x^2 - 6x + 9 = 0$ (b) $x^2 - 10x + 21 = 0$ (c) $x^2 - 14x + 49 = 0$ (d) none of these

Q 46. If $[.]$ denotes the greatest integer function then $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2}$ is

- (a) 0 (b) x (c) $\frac{x}{2}$ (d) $\frac{x^2}{2}$

Q 47. $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$ is equal to

- (a) 1 (b) 2 (c) 0 (d) none of these

Q 48. $\lim_{x \rightarrow a} \frac{x}{x-a} \cdot \int_a^x f(x) dx$ is equal to

- (a) $f(a)$ (b) $af(a)$ (c) 0 (d) none of these

Q 49. $\lim_{x \rightarrow 1+0} \frac{\int_1^x |t-1| dt}{\sin(x-1)}$ is equal to

- (a) 0 (b) 1 (c) -1 (d) none of these

Q 50. Let $f(x) = \sin x, x \neq nx$

$2, x = 2\pi, \text{ where } n \in \mathbb{Z},$

Then $\lim_{x \rightarrow 0} g(f(x))$ is

- (a) 0 (b) 1 (c) 3 (d) none of these

Q 51. If $f(x)$ continuous in $[0, 1]$ and $f\left(\frac{1}{3}\right) = 1$ then $\lim_{n \rightarrow \infty} f\left(\frac{n}{\sqrt{9n^2 + 1}}\right)$ is

- (a) 1 (b) 0 (c) $\frac{1}{3}$ (d) none of these

Q 52. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$ then $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is equal to

- (a) $\frac{9}{2}$ (b) $\frac{2}{9}$ (c) 0 (d) none of these

Choose the correct options. One or more options may be correct.

Q 53. Let $f(x) = 1 + \frac{2x}{a}$, $0 \leq x < 1$

ax , $1 \leq x < 2$.

If $\lim_{x \rightarrow 0} f(x)$ exists then a is

- (a) 1 (b) -1 (c) 2 (d) -2

Q 54. If α is a repeated root of $ax^2 + bx + c = 0$ then $\lim_{x \rightarrow \alpha} \frac{\sin(ax^2 + bx + c)}{(x - \alpha)^2}$ is

- (a) 0 (b) a (c) b (d) c

Q 55. If $f(x) = |x - 1| - [x]$ = the greatest integer less than or equal to x , then

- (a) $f(1 + 0) = -1$, $f(1 - 0) = 0$ (b) $f(1 + 0) = 0 = -f(1 - 0)$
(c) $\lim_{x \rightarrow 1} f(x)$ exists (d) $\lim_{x \rightarrow 1} f(x)$ does not exist

Q 56. If $\lim_{n \rightarrow \infty} \left(an - \frac{1 + n^2}{1 + n} \right) = b$, a finite number, then

- (a) $a = 1$ (b) $a = 0$ (c) $b = 1$ (d) $b = -1$

Q 57. Let $\tan \alpha \cdot x + \sin \alpha \cdot y = \alpha$ and $\alpha \operatorname{cosec} \alpha \cdot x + \cos \alpha \cdot y = 1$ be two variable straight lines, α being the parameter. Let P be the point of intersection of the lines. In the limiting position when $\alpha \rightarrow 0$, the point P lies on the line

- (a) $x = 2$ (b) $x = -1$ (c) $y + 1 = 0$ (d) $y = 2$

Answers

1b	2a	3c	4b	5a	6c	7b	8d	9d	10a
11a	12c	13c	14a	15b	16a	17c	18c	19b	20a
21b	22a	23c	24a	25b	26c	27a	28c	29c	30b
31d	32d	33b	34c	35c	36b	37a	38a	39d	40c
41c	42c	43c	44d	45b	46c	47a	48b	49a	50b
51a	52b	53bc	54b	55ad	56ac	57ac			

Continuity and Differentiability

Choose the most appropriate option (a, b, c or d).

- Q 1. Let $f(x) = \log |x - 1|$, $x \neq 1$. The value of $f'\left(\frac{1}{2}\right)$
- (a) is -2 (b) is 2 (c) does not exist (d) none of these
- Q 2. Let $y = \left| \tan\left(\frac{\pi}{4} - x\right) \right|$. Then $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$
- (a) is 1 (b) is -1 (c) does not exist (d) none of these
- Q 3. Let $y = |x| + |x - 2|$. Then $\frac{dy}{dx}$ at $x = 2$
- (a) is 2 (b) is 0 (c) does not exist (d) none of these
- Q 4. Let $f(x) = \lambda + \mu |x| + \nu |x|^2$, where λ, μ, ν are real constants. The $f'(0)$ exists if
- (a) $\mu = 0$ (b) $\nu = 0$ (c) $\lambda = 0$ (d) $\mu = \nu$
- Q 5. If $f(x) = \frac{[x]}{|x|}$, $x \neq 0$ where $[.]$ denotes the greatest integer function, then $f'(1)$ is
- (a) -1 (b) ∞ (c) nonexistent (d) none of these
- Q 6. If $f(x) = |\cos 2x|$ then $f'\left(\frac{\pi}{4} + 0\right)$ is equal to
- (a) 2 (b) 0 (c) -2 (d) none of these
- Q 7. If $(x) = \sin \pi[x]$ then $f'(1 - 0)$ is equal to
- (a) -1 (b) 0 (c) 1 (d) none of these
- Q 8. Let $f(x) = [x^2] - [x]^2$, where $[.]$ denotes the greatest integer function. Then
- (a) $f(x)$ is discontinuous for all integral values of x
(b) $f(x)$ is discontinuous only at $x = 0, 1$
(c) $f(x)$ is continuous only at $x = 1$ (d) none of these
- Q 9. Let $f(x) = [\cos x + \sin x]$, $0 < x < 2\pi$ where $[x]$ denotes the greatest integer less than or equal to x . The number of points of discontinuity of $f(x)$ is
- (a) 6 (b) 5 (c) 4 (d) 3

- Q 10. Let $f(x) = x - |x - x^2|$, $x \in [-1, 1]$. Then the number of points at which $f(x)$ is discontinuous is
- (a) 1 (b) 2 (c) 0 (d) none of these

- Q 11. Let $f(x) = \sqrt{1+x^2}$, $x < \sqrt{3}$

$$\sqrt{3}x - 1, \sqrt{3} \leq x < 4$$

$$[x], 4 \leq x < 5, \text{ where } [x] \text{ is the greatest integer } \leq x$$

$$|1-x|, x \geq 5$$

The number of points of discontinuity of $f(x)$ in $\mathbb{R}$ is

- (a) 3 (b) 0 (c) infinite (d) none of these
- Q 12. Let $f(x) = \int_0^x t \sin \frac{1}{t} dt$. Then the number of points of discontinuity of the function $f(x)$ in the open interval $(0, \pi)$ is

- (a) 0 (b) 1 (c) 2 (d) infinite

- Q 13. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Then

- (a) $f(x) = x$ for all least one $0 \leq x \leq 1$ (b) $f(x)$ will be differentiable in $[0, 1]$
- (c) $f(x) + x = 0$ for at least one x such that $0 \leq x \leq 1$ (d) none of these

- Q 14. Let $f(x)$ be a continuous function defined for $1 \leq x \leq 3$. If $f(x)$ takes rational values for all x and $f(2) = 10$ then the value of $f(1.5)$ is

- (a) 7.5 (b) 10 (c) 5 (d) none of these

- Q 15. If $f(x) = e^{-1/x^2}$, $x \neq 0$, and $f(0) = 0$ then $f'(0)$ is

- (a) 0 (b) 1 (c) e (d) nonexistent

- Q 16. Let $f(x) = \sin x$, $g(x) = [x + 1]$ and $g\{f(x)\} = h(x)$, where $[.]$ is the greatest integer function. Then $h'\left(\frac{\pi}{2}\right)$ is

- (a) nonexistent (b) 1 (c) -1 (d) none of these

- Q 17. Let $f(x) = [x]$, $g(x) = |x|$ and $f\{g(x)\} = h(x)$, where $[.]$ is the greatest integer function. Then $h'(-1)$ is

- (a) 0 (b) $-\infty$ (c) nonexistent (d) none of these

Q 18. The number of values of $x \in [0, 2]$ at which the real function $f(x) = \left|x - \frac{1}{2}\right| + |x - 1| + \tan x$ is not finitely differentiable is

- (a) 2 (b) 3 (c) 1 (d) 0

Q 19. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in \mathbb{Z}$, p is a prime number and $[x]$ = the greatest integer less than or equal to x . The number of points at which $g(x)$ is not differentiable is

- (a) p (b) $p - 1$ (c) $2p + 1$ (d) $2p - 1$

Q 20. Let $f(x) = (x - 1)^2 \cos \frac{1}{x - 1} - |x|$, $x \neq 1$

$$-1, x = 1$$

The set of points where $f(x)$ is not differentiable is

- (a) $\{1\}$ (b) $\{0, 1\}$ (c) $\{0\}$ (d) none of these

Q 21. Let $f(x) = \frac{1 - \sin x}{\sin 2x}$, $x \neq \frac{\pi}{2}$. If $f(x)$ is continuous at $x = \frac{\pi}{2}$ then $f\left(\frac{\pi}{2}\right)$ should be

- (a) 1 (b) 0 (c) $\frac{1}{2}$ (d) none of these

Q 22. A function $f(x)$ is defined as below

$$f(x) = \frac{\cos(\sin x) - \cos x}{x^2}, x \neq 0 \text{ and } f(0) = a.$$

$f(x)$ is continuous at $x = 0$ if a equals

- (a) 0 (b) 4 (c) 5 (d) 6

Q 23. Let $f(x) = \frac{1 - \tan x}{4x - \pi}$, $x \neq \frac{\pi}{4}$ and $x \in \left[0, \frac{\pi}{2}\right)$

$$\lambda, x = \frac{\pi}{4}$$

If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right)$ then λ is

- (a) 1 (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) none of these

Q 24. Let $f(x) = (\sin x)^{\frac{1}{\pi - 2x}}$, $x \neq \frac{\pi}{2}$.

If $f(x)$ is continuous at $x = \frac{\pi}{2}$ then $f\left(\frac{\pi}{2}\right)$ is

- (a) e (b) 1 (c) 0 (d) none of these

Q 25. Let $f(x) = \sin \frac{1}{x}$, $x \neq 0$. Then $f(x)$ can be continuous at $x = 0$

- (a) if $f(0) = 1$ (b) if $f(0) = 0$ (c) if $f(0) = -1$ (d) for no value of $f(0)$

Q 26. If $f(x) = px^2 - q$, $x \in [0, 1)$

$$x + 1, x \in (1, 2]$$

and $f(1) = 2$ then the value of the pair (p, q) for which $f(x)$ cannot be continuous at $x = 1$ is

- (a) $(2, 0)$ (b) $(1, -1)$ (c) $(4, 2)$ (d) $(1, 1)$

Q 27. If $f(x) = x$, $x \leq 1$, and $f(x) = x^2 + bx + c$, $x > 1$, and $f'(x)$ exists finitely for all $x \in \mathbb{R}$ then

- (a) $b = -1, c \in \mathbb{R}$ (b) $c = 1, b \in \mathbb{R}$ (c) $b = 1, c = -1$ (d) $b = -1, c = 1$

Choose the correct options. One or more options may be correct.

Q 28. If $f(x) = e^x$, $x < 2$

$$a + bx, x \geq 2$$

is differentiable for all $x \in \mathbb{R}$ then

- (a) $a + b = 0$ (b) $a + 2b = e^2$ (c) $b = e^2$ (d) none of these

Q 29. If $f(x) = \cos^{-1}(\cos x)$ then $f(x)$ is

- (a) continuous at $x = \pi$ (b) discontinuous at $x = -\pi$
(c) differentiable at $x = 0$ (d) nondifferentiable at $x = \pi$

Q 30. Let $f(x) = x - |x|$. Then

- (a) $f(x)$ is continuous everywhere (b) $f(x)$ is differentiable everywhere
(c) $f(x)$ is discontinuous at $x = 0$ (d) $f(x)$ is not differentiable at $x = 0$

Q 31. If $f(x) = [x] + \left[x + \frac{1}{2}\right]$, where $[.]$ denotes the greatest integer function, then

- (a) $f(x)$ is continuous at $x = \frac{1}{2}$ (b) $\lim_{x \rightarrow 1/2+0} f(x) = 1$

(c) $f(x)$ is discontinuous at $x = \frac{1}{2}$

(d) $\lim_{x \rightarrow 1/2-0} f(x) = 1$

Q 32. If $f(x) = |2 - x| + (2 + x)$, where $(x) =$ the least integer greater than or equal to x , then

(a) $f(2 - 0) = f(2) = 4$

(b) $f(x)$ is continuous at $x = 2$

(c) $f(x)$ is nondifferentiable at $x = 2$

(d) $f(x)$ is differentiable but not continuous at $x = 2$

Q 33. Let $h(x) = \min \{x, x^2\}$ for every real number x . Then

(a) h is continuous for all x

(b) h is differentiable for all x

(c) $h'(x) = 1$ for all $x > 1$

(d) h is not differentiable at two values of x

Q 34. At $x = 0$, the function $y = e^{-|x|}$ is

(a) continuous

(b) continuous and differentiable

(c) differentiable with derivative = 1

(d) differentiable with derivative = -1

Q 35. A function $f(x)$ is defined as follows :

$$f(x) = -x^2, x \leq 0 \quad f(x) = 5x - 4, 0 < x \leq 1$$

$$f(x) = 4x^2 - 3x, 1 < x \leq 2 \quad f(x) = 3x + 4, x > 2$$

(a) $f(x)$ is not continuous at $x = 0$, but differentiable there

(b) $f(x)$ is continuous at $x = 1$, but not differentiable there

(c) $f(x)$ is continuous at $x = 2$, but not differentiable there

(d) none of the above

Q 36. The function $f(x) = \frac{1}{x} - \frac{2}{e^{2x} - 1}$, $x \neq 0$, is continuous at $x = 0$. Then

(a) $f(0) = 1$

(b) $f(x)$ is differentiable at $x = 0$

(c) $f(x)$ is not differentiable at $x = 0$

(d) $f'(0) = \frac{1}{3}$

Q 37. The function $f(x) = |x^2 - 3x + 2| + \cos |x|$ is not differentiable at $x =$

(a) -1

(b) 0

(c) 1

(d) 2

Q 38. Let $f(x)$ be defined as follows :

$$f(x) = x^6, x^2 > 1$$

$$x^3, x^2 \leq 1$$

Then $f(x)$ is

- (a) continuous everywhere (b) differentiable everywhere
(c) discontinuous at $x = -1$ (d) not differentiable at $x = 1$

Q 39. Let $f(x) = \sin x, x \geq 0$

$$-\sin x, x < 0$$

The $f(x)$ is

- (a) continuous at $x = 0$ (b) differentiable at $x = 0$
(c) discontinuous at $x = 0$ (d) not differentiable at $x = 0$

Q 40. If $f(x) = \sum_{n=0}^n a_n |x|^n$, where a_i 's real constants, then $f(x)$ is

- (a) continuous at $x = 0$ for all a_i (b) differentiable at $x = 0$ for all $a_i \in \mathbb{R}$
(c) differentiable at $x = 0$ for $a_{2k+1} = 0$ (d) none of these

Q 41. Let $[x]$ denote the greatest integer less than or equal to x . Now $g(x)$ is defined as below :

$$g(x) = [f(x)], x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$$

$$= 3, x = \frac{\pi}{2}$$

where $f(x) = \frac{2(\sin x - \sin^n x) + |\sin x - \sin^n x|}{2(\sin x - \sin^n x) - |\sin x - \sin^n x|}$, $n \in \mathbb{R}$. Then

- (a) $g(x)$ is continuous and differentiable at $x = \frac{\pi}{2}$ when $n > 1$
(b) $g(x)$ is continuous and differentiable at $x = \frac{\pi}{2}$ when $0 < n < 1$
(c) $g(x)$ is continuous but not differentiable at $x = \frac{\pi}{2}$ when $n > 1$
(d) $g(x)$ is continuous but differentiable at $x = \frac{\pi}{2}$ when $0 < n < 1$

Q 42. Let $f(x) = \phi(x) + \psi(x)$ and $\phi'(a), \psi'(a)$ are finite and definite. Then

(a) $f(x)$ is continuous at $x = a$

(b) $f(x)$ is differentiable at $x = a$

(c) $f(x)$ is continuous at $x = a$

(d) $f'(x)$ is differentiable at $x = a$

Q 43. Let $f(x) = x + |x|$. Then $f(x)$ is

(a) differentiable at all x

(b) continuous at all x

(c) differentiable everywhere except at $x = 0$

(d) continuous everywhere except at $x = 0$

Q 44. Let $f(x) = \lim_{n \rightarrow \infty} \frac{1 - x^n}{1 + x^n}$. Then

(a) $f(x)$ is a constant in $0 < x < 1$

(b) $f(x)$ is continuous at $x = 1$

(c) $f(x)$ is not differentiable at $x = 1$

(d) none of these

Q 45. Let $f(x) = 1 - |\cos x|$ for all $x \in \mathbb{R}$. Then

(a) $f'\left(\frac{\pi}{2}\right)$ does not exist

(b) $f(x)$ is continuous everywhere

(c) $f(x)$ is not differentiable anywhere

(d) $\lim_{x \rightarrow \pi/2+0} f(x) = 1$

Q 46. Let $f(x) = [\tan^2 x]$, where $[.]$ denotes the greatest integer function. Then

(a) $\lim_{x \rightarrow 0} f(x)$ does not exist

(b) $f(x)$ is continuous at $x = 0$

(c) $f'(0) = 1$

(d) $f(x)$ is not differentiable at $x = 0$

Q 47. If $f(x) = \frac{x}{\sqrt{x+1} - \sqrt{x}}$ be a real-valued function then

(a) $f(x)$ is continuous, but $f'(0)$ does not exist

(b) $f(x)$ is differentiable at $x = 0$

(c) $f(x)$ is not continuous at $x = 0$

(d) $f(x)$ is not differentiable at $x = 0$

Q 48. A function $f(x)$ is defined in the interval $[1, 4)$ as follows :

$$f(x) = \log_e[x], 1 \leq x < 3$$

$$|\log_e x|, 3 \leq x < 4$$

The graph of the function $f(x)$

(a) is broken at two points

(b) is broken at exactly one point

(c) does not have a definite tangent at two points

(d) does not have a definite tangent at more than two points

1a	2c	3c	4a	5c	6a	7b	8c	9c	10c
11d	12a	13a	14b	15c	16a	17c	18b	19d	20c
21b	22a	23c	24b	25d	26d	27d	28abc	29ad	30ad
31ab	32ac	33acd	34a	35c	36ab	37cd	38cd	39ad	40ac
41b	42ab	43bc	44ac	45bd	46b	47b	48ac		

Application of dy/dx

Choose the most appropriate option (a, b, c or d).

- Q 1. If m be the slope of tangent to the curve $e^y = 1 + x^2$ then
(a) $|m| > 1$ (b) $m < 1$ (c) $|m| < 1$ (d) $|m| \leq 1$
- Q 2. If at each point of the curve $y = x^3 - ax^2 + x + 1$ the tangent is inclined at an acute angle with the positive direction of the x -axis then
(a) $a > 0$ (b) $a \leq \sqrt{3}$ (c) $-\sqrt{3} \leq a \leq \sqrt{3}$ (d) none of these
- Q 3. The slope of the tangent to the curve $y = x^2 - x$ at the point where the line $y = 2$ cuts the curve in the first quadrant is
(a) 2 (b) 3 (c) -3 (d) none of these
- Q 4. The slope of the tangent to the curve $y = \sqrt{4 - x^2}$ at the point where the ordinate the abscissa are equal, is
(a) -1 (b) 1 (c) 0 (d) none of these
- Q 5. The slope of the tangent to the locus $y = \cos^{-1}(\cos x)$ at $x = -\frac{\pi}{4}$ is
(a) 1 (b) 0 (c) 2 (d) -1
- Q 6. The slope of the tangent to the curve $y = \int_0^x \frac{dx}{1+x^3}$ at the point where $x = 1$ is
(a) $\frac{1}{2}$ (b) 1 (c) $\frac{1}{4}$ (d) none of these
- Q 7. The equation of the curve is given by $x = e^t \sin t$, $y = e^t \sin t$. The inclination of the tangent to the curve at the point $t = \frac{\pi}{4}$ is
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) 0
- Q 8. The curve given by $x + y = e^{xy}$ has a tangent parallel to the y -axis at the point
(a) (0, 1) (b) (1, 0) (c) (1, 1) (d) none of these
- Q 9. $P(2, 2)$ and $Q\left(\frac{1}{2}, -1\right)$ are two points on the parabola $y^2 = 2x$. The coordinates of the point R on the parabola, where the tangent to the curve is parallel to the chord PQ, is

- (a) $\left(\frac{5}{4}, \sqrt{\frac{5}{2}}\right)$ (b) (2, -1) (c) $\left(\frac{1}{8}, \frac{1}{2}\right)$ (d) none of these

Q 10. The number of tangent to the curve $x^{3/2} + y^{3/2} = a^{3/2}$, where the tangents are equally inclined to the axes, is

- (a) 2 (b) 1 (c) 0 (d) 4

Q 11. The point on the curve $\sqrt{x} + \sqrt{y} = 2a^2$, where the tangent is equally inclined to the axes, is

- (a) (a^4, a^4) (b) $(0, 4a^4)$ (c) $(4a^4, 0)$ (d) none of these

Q 12. The parabola $x^2 = 5 - 4y$ and $y = x^2$ cut at the point (1, 1) at an angle

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) none of these

Q 13. The angle between two tangents to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the points where the line $y = 1$ cuts the curve is

- (a) $\frac{\pi}{4}$ (b) $\tan^{-1} \frac{6\sqrt{2}}{7}$ (c) $\frac{\pi}{2}$ (d) none of these

Q 14. The number of tangents to the curve $y^2 - 2x^3 - 4y + 8 = 0$ that pass through (1, 2) is

- (a) 3 (b) 1 (c) 2 (d) 6

Q 15. The equation of a curve is $y = f(x)$. The tangents at (1, f(1)), (2, f(2)) and (3, f(3)) make angles

$\frac{\pi}{6}, \frac{\pi}{3}$ and $\frac{\pi}{4}$ respectively with the positive direction of the x-axis. Then the value of

$\int_2^3 f(x)f''(x)dx + \int_1^3 f''(x)dx$ is equal to

- (a) $-\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}$ (c) 0 (d) none of these

Q 16. The equation of the tangent to the curve $y = e^{-|x|}$ at the point where the curve cuts the line $x = 1$ is

- (a) $x + y = e$ (b) $e(x + y) = 1$ (c) $y + ex = 1$ (d) none of these

Q 17. The equation of the tangent to the curve $y = bx^{-x/a}$ where it cuts the y-axis is

- (a) $\frac{x}{a} + \frac{y}{b} = 1$ (b) $\frac{x}{a} + \frac{y}{b} = -1$ (c) $\frac{x}{a} - \frac{y}{b} = 1$ (d) none of these

- Q 18. If the tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at any point on it cuts the axes OX and OY at P and Q respectively then OP + OQ is
- (a) 2a (b) a (c) $\frac{1}{2}a$ (d) none of these
- Q 19. If the line joining the point (0, 3) and (5, -2) is a tangent to the curve $y = \frac{c}{x+1}$ then the value of c is
- (a) 1 (b) -2 (c) 4 (d) none of these
- Q 20. The curve $\frac{x^n}{a^n} + \frac{y^n}{b^n} = 2$ touches the line $\frac{x}{a} + \frac{y}{b} = 2$ at the point
- (a) (b, a) (b) (a, b) (c) (1, 1) (d) $\left(\frac{1}{b}, \frac{1}{a}\right)$
- Q 21. The sum of the intercepts made on the axes of coordinates by any tangent to the curve $\sqrt{x} + \sqrt{y} = 2$ is equal to
- (a) 4 (b) 2 (c) 8 (d) none of these
- Q 22. The area bounded by the axes of reference and the normal to $y = \log_e x$ at the point (1, 0) is
- (a) 1 unit² (b) 2 unit² (c) $\frac{1}{2}$ unit² (d) none of these
- Q 23. The normal to the curve $2x^2 + y^2 = 12$ at the point (2, 2) cuts the curve again at
- (a) $\left(-\frac{22}{9}, -\frac{2}{9}\right)$ (b) $\left(\frac{22}{9}, \frac{2}{9}\right)$ (c) (-2, -2) (d) none of these
- Q 24. Two cyclists start from the junction of two perpendicular roads, their velocities being 3v metres/minute and 4v metres/minute. The rate at which the two cyclists are separating is
- (a) $\frac{7}{2}v$ m/min (b) 5v m/min (c) v m/min (d) none of these
- Q 25. A stick of length a cm rests against a vertical wall and the horizontal floor. If the foot of the stick slides with a constant velocity of b cm/s then the magnitude of the velocity of the middle point of the stick when it is equally inclined with the floor and the wall, is
- (a) $\frac{b}{\sqrt{2}}$ cm/s (b) $\frac{b}{2}$ cm/s (c) $\frac{ab}{2}$ cm/s (d) none of these

- Q 26. If $y = \int_0^x \frac{t^2}{\sqrt{t^2 + 1}} dt$ then rate of change of y with respect to x when $x = 1$, is
- (a) $\sqrt{2}$ (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) none of these
- Q 27. On the curve $x^3 = 12y$ the abscissa change at a faster rate than the ordinate. Then x belongs to the interval
- (a) $(-2, 2)$ (b) $(-1, 1)$ (c) $(0, 2)$ (d) none of these
- Q 28. A balloon is pumped at the rate of $a \text{ cm}^3/\text{minute}$. The rate of increase of its surface area when the radius is $b \text{ cm}$, is
- (a) $\frac{2a^2}{b^4} \text{ cm}^2/\text{min}$ (b) $\frac{a}{2b} \text{ cm}^2/\text{min}$ (c) $\frac{2a}{b} \text{ cm}^2/\text{min}$ (d) none of these
- Q 29. x and y are the sides of two squares such that $y = x - x^2$. The rate of change of the area of the second square respect to that of the first square is
- (a) $2(1 - x^2)x$ (b) $2x^2 - 3x + 1$ (c) $2(2x^2 - 3x + 1)$ (d) none of these
- Q 30. Let the equation of a curve be $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$. If θ changes at a constant rate k then the rate of change of the slope of the tangent to the curve at $\theta = \frac{\pi}{3}$ is
- (a) $\frac{2k}{\sqrt{3}}$ (b) $\frac{k}{\sqrt{3}}$ (c) k (d) none of these
- Q 31. If there is an error of k% in measuring the edge of a cube then the per cent error in estimating its volume is
- (a) k (b) 3k (c) $\frac{k}{3}$ (d) none of these
- Q 32. If $1^\circ = \alpha$ radius then the approximate value of $\cos 60^\circ 1'$ is
- (a) $\frac{1}{2} + \frac{\alpha\sqrt{3}}{120}$ (b) $\frac{1}{2} - \frac{\alpha}{120}$ (c) $\frac{1}{2} - \frac{\alpha\sqrt{3}}{120}$ (d) none of these

Choose the correct options. One or more options may be correct.

- Q 33. Let the parabola $y = x^2 + ax + b$ and $y = x(c - x)$ touches each other at the point $(1, 0)$. Then
- (a) $a = -3$ (b) $b = 1$ (c) $c = 2$ (d) $b + c = 3$

Q 34. Let $y = f(x)$ be the equation of a parabola which is touched by the line $y = x$ at the point where $x = 1$. Then

- (a) $f'(0) = f'(1)$ (b) $f'(1) = 1$ (c) $f(0) + f'(0) + f''(0) = 1$ (d) $2f(0) = 1 - f'(0)$

Q 35. A point on the ellipse $4x^2 + 9y^2 = 36$ where the tangent is equally inclined to the axes is

- (a) $\left(\frac{9}{\sqrt{13}}, \frac{4}{\sqrt{13}}\right)$ (b) $\left(-\frac{9}{\sqrt{13}}, \frac{4}{\sqrt{13}}\right)$ (c) $\left(\frac{9}{\sqrt{13}}, -\frac{4}{\sqrt{13}}\right)$ (d) none of these

Q 36. If the line $ax + by + c = 0$ is a normal to the rectangular hyperbola $xy = 1$ then

- (a) $a > 0, b > 0$ (b) $a > 0, b < 0$ (c) $a < 0, b > 0$ (d) $a < 0, b < 0$

Q 37. A tangent to the curve $y = \int_0^x |t| dt$, which is parallel to the line $y = x$, cuts off an intercept from the y -axis equal to

- (a) 1 (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) -1

1d 2c 3b 4a 5d 6a 7d 8b 9c 10b

11a 12a 13b 14c 15a 16d 17a 18b 19c 20b

21a 22c 23a 24b 25a 26c 27a 28c 29b 30d

31b 32c 33ad 34bd 35abc 36bc 37bc

Maxima and Minima

- Q 1. If $xy = a^2$ and $S = b^2x + c^2y$ where a , b and c are constants then the minimum value of S is
- (a) abc (b) $bc\sqrt{a}$ (c) $2abc$ (d) none of these
- Q 2. The global minimum value of $f(x) = x^4 - x^2 - 2x + 6$ is
- (a) 6 (b) 8 (c) 4 (d) nonexistent
- Q 3. The maximum value of $f(x) = 3 \cos^2 x + 4 \sin^2 x + \cos \frac{x}{2} + \sin \frac{x}{2}$ is
- (a) 4 (b) $3 + \sqrt{2}$ (c) $4 + \sqrt{2}$ (d) none of these
- Q 4. If $a > b > 0$, the minimum value of $a \sec \theta - b \tan \theta$ is
- (a) $b - a$ (b) $\sqrt{a^2 + b^2}$ (c) $\sqrt{a^2 - b^2}$ (d) $2\sqrt{a^2 - b^2}$
- Q 5. The number of values of x where the function $f(x) = \cos x + \cos(\sqrt{2}x)$ attains its maximum is
- (a) 0 (b) 1 (c) 2 (d) infinite
- Q 6. The function $f(x) = \frac{x}{1 + x \tan x}$ has
- (a) one point of minimum in the interval $(0, \pi/2)$
(b) one point of maximum in the interval $(0, \pi/2)$
(c) no point of maximum, no point of minimum in the interval $(0, \pi/2)$
(d) two points of maxima in the interval $(0, \pi/2)$
- Q 7. Let $f(x) = e^x \sin x$ be the equation of a curve. If at $x = a$, $0 \leq a \leq 2\pi$, the slope of the tangent is the maximum then the value of a is
- (a) $\pi/2$ (b) $3\pi/2$ (c) π (d) $\pi/4$
- Q 8. Let the tangent to the graph of $y = f(x)$ at the point $x = a$ be parallel to the x -axis, and let $f'(a - h) > 0$ and $f'(a + h) < 0$, where h is a very small positive number. Then the ordinate of the point is
- (a) a maximum (b) a minimum
(c) both a maximum and a minimum (d) neither a maximum nor a minimum
- Q 9. Let $f(x) = x^3 + 3x^2 - 9x + 2$. Then
- (a) $f(x)$ has a maximum at $x = 1$
(b) $f(x)$ has neither a minimum nor a maximum at $x = -3$

- (c) $f(x)$ has a minimum at $x = 1$ (d) none of these
- Q 10. If $f(x) = a \log_e |x| + bx^2 + x$ has extremums at $x = 1$ and $x = 3$ then
- (a) $a = -\frac{3}{4}, b = -\frac{1}{8}$ (b) $a = \frac{3}{4}, b = -\frac{1}{8}$ (c) $a = -\frac{3}{4}, b = \frac{1}{8}$ (d) none of these
- Q 11. The maximum value of $\left(\frac{1}{x}\right)^{2x^2}$ is
- (a) e (b) $\sqrt[e]{e}$ (c) 1 (d) none of these
- Q 12. The maximum ordinate of a point on the graph of the function $f(x) = \sin x(1 + \cos x)$ is
- (a) $\frac{2 + \sqrt{3}}{4}$ (b) $\frac{3\sqrt{3}}{4}$ (c) 1 (d) none of these
- Q 13. If $\theta + \phi = \frac{\pi}{3}$ then $\sin \theta \cdot \sin \phi$ has a maximum value at $\theta =$
- (a) $\frac{\pi}{6}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{4}$ (d) none of these
- Q 14. Let $f(x) = x^3 - 6x^2 + 12x - 3$. Then at $x = 2$, $f(x)$ has
- (a) a maximum (b) a minimum
- (c) both a maximum and a minimum (d) neither a maximum nor a minimum
- Q 15. Let $f(x) = (x - p)^2 + (x - q)^2 + (x - r)^2$. Then $f(x)$ has a minimum at $x = \lambda$, where λ is equal to
- (a) $\frac{p+q+r}{3}$ (b) $\sqrt[3]{pqr}$ (c) $\frac{3}{\frac{1}{p} + \frac{1}{q} + \frac{1}{r}}$ (d) none of these
- Q 16. Let $f(x) = 1 + 2x^2 + 2^2x^4 + \dots + 2^{10}x^{20}$. Then $f(x)$ has
- (a) more than one minimum (b) exactly one minimum
- (c) at least one maximum (d) none of these
- Q 17. Let $f(x) = \frac{a}{x} + x^2$. If it has a maximum at $x = -3$ then a is
- (a) -1 (b) 16 (c) 1 (d) none of these
- Q 18. Let $f(x)$ be a function such that $f'(a) \neq 0$. Then at $x = a$, $f(x)$
- (a) cannot have a maximum (b) cannot have a minimum

(c) must have neither a maximum nor a minimum (d) none of these

Q 19. Let the function $f(x)$ be defined as below.

$$f(x) = \sin^{-1}\lambda + x^2, 0 < x < 1$$

$$2x, x \geq 1$$

$f(x)$ can have a minimum at $x = 1$ if the value of λ is

(a) 1 (b) -1 (c) 0 (d) none of these

Q 20. Let x be a number which exceeds its square by the greatest possible quantity. Then x is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) none of these

Q 21. The sum of two nonzero numbers is 8. The minimum value of the sum of their reciprocals is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) none of these

Q 22. In a $\triangle ABC$, $\angle B = 90^\circ$ and $b + a = 4$. The area of the triangle is the maximum when $\angle C$ is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) none of these

Q 23. The point $(0, 3)$ is nearest to the curve $x^2 = 2y$ at

(a) $(2\sqrt{2}, 0)$ (b) $(0, 0)$ (c) $(2, 2)$ (d) none of these

Q 24. If λ, μ be real numbers such that $x^3 - \lambda x^2 + \mu x - 6 = 0$ has its roots real and positive then the minimum value of μ is

(a) $3 \times \sqrt[3]{36}$ (b) 11 (c) 0 (d) none of these

Choose the correct options. One or more options may be correct.

Q 25. Let $f(x) = x + x^{-1}$. Then

(a) $f(x)$ has a maximum but no minimum (b) $f(x)$ has no maximum but a minimum
(c) $f(x)$ has a maximum and a minimum (d) $\max f(x) < \min f(x)$

Q 26. Let $f(x) = ax^3 + bx^2 + cx + 1$ have extrema at $x = \alpha, \beta$ such that $\alpha\beta < 0$ and $f(\alpha), f(\beta) < 0$. Then the equation $f(x) = 0$ has

(a) three equal real roots (b) three distinct real roots
(c) one positive root if $f(\alpha) < 0$ and $f(\beta) > 0$ (d) one negative root if $f(\alpha) > 0$ and $f(\beta) < 0$

Q 27. The critical point(s) of $f(x) = \frac{|2-x|}{x^2}$ is (are)

- (a) $x = 0$ (b) $x = 2$ (c) $x = 4$ (d) none of these

Q 28. The value of x for which the function $f(x) = \int_0^x (1-t^2)e^{-t^2/2} dt$ has an extremum is

- (a) 0 (b) 1 (c) -1 (d) none of these

Q 29. Let $f(x) = x^3 + 3x^2 + 2x + 2$. Then, at $x = -1$

- (a) $f(x)$ has a maximum (b) $f(x)$ has a minimum (c) $f'(x)$ has a maximum (d) $f'(x)$ has a minimum

Q 30. The function $f(x) = x^2 + \frac{\lambda}{x}$ has a

- (a) minimum at $x = 2$ if $\lambda = 16$ (b) maximum at $x = 2$ if $\lambda = 16$
(c) maximum for no real value of λ (d) point of inflection at $x = 1$ if $\lambda = -1$

Q 31. Let $f(x) = (x-1)^4 \cdot (x-2)^n$, $n \in \mathbb{N}$. Then $f(x)$ has

- (a) a maximum at $x = 1$ if n is odd (b) a maximum $x = 1$ if n is even
(c) a minimum at $x = 2$ if n is even (d) a maximum at $x = 2$ if n is odd

1c 2c 3c 4c 5b 6b 7a 8a 9c 10a
11b 12b 13a 14d 15a 16b 17d 18d 19d 20a
21b 22c 23c 24a 25cd 26bcd 27abc 28bc 29d 30acd
31ac

Monotonic Functions and Lagrange's Theorem

Choose the most appropriate option (a, b, c or d).

- Q 1. If $f(x) = x^3 + 4x^2 + \lambda x + 1$ is a monotonically decreasing function of x in the largest possible interval $(-2, -2/3)$ then
- (a) $\lambda = 4$ (b) $\lambda = 2$ (c) $\lambda = -1$ (d) λ has no real value
- Q 2. The function $f(x) = \sin^4 x + \cos^4 x$ increases if
- (a) $0 < x < \frac{\pi}{8}$ (b) $\frac{\pi}{4} < x < \frac{3\pi}{8}$ (c) $\frac{3\pi}{4} < x < \frac{5\pi}{8}$ (d) $\frac{5\pi}{8} < x < \frac{3\pi}{4}$
- Q 3. If $f(x) = \frac{x}{\sin x}$ and $g(x) = \frac{x}{\tan x}$, where $0 < x \leq 1$, then in the interval
- (a) both $f(x)$ and $g(x)$ are increasing functions (b) both $f(x)$ and $g(x)$ are decreasing functions
- (c) $f(x)$ is an increasing function (d) $g(x)$ is an increasing function
- Q 4. The function $f(x) = \tan^{-1} x - x$ is monotonically decreasing in the set
- (a) $\mathbb{R}$ (b) $(0, +\infty)$ (c) $\mathbb{R} - \{0\}$ (d) none of these
- Q 5. Let $f(x) = \tan^{-1}\{\phi(x)\}$, where $\phi(x)$ is m.i. for $0 < x < \pi/2$. Then $f(x)$ is
- (a) increasing in $(0, \pi/2)$ (b) decreasing in $(0, \pi/2)$
- (c) increasing in $(0, \pi/4)$ and decreasing in $(\pi/4, \pi/2)$ (d) none of these
- Q 6. If $f(x) = \frac{p^2 - 1}{p^2 + 1} x^3 - 3x + \log 2$ is a decreasing function of x in $\mathbb{R}$ then the set of possible values of p (independent of x) is
- (a) $[-1, 1]$ (b) $[1, \infty)$ (c) $(-\infty, -1]$ (d) none of these
- Q 7. If $f(x) = (ab - b^2 - 2)x + \int_0^x (\cos^4 \theta + \sin^4 \theta) d\theta$ is a decreasing function of x for all $x \in \mathbb{R}$ and $b \in \mathbb{R}$, b being independent of x , then
- (a) $a \in (0, \sqrt{6})$ (b) $a \in (-\sqrt{6}, \sqrt{6})$ (c) $a \in (-\sqrt{6}, 0)$ (d) none of these
- Q 8. Let $f(x) = \cos \pi x + 10x + 3x^2 + x^3$, $-2 \leq x \leq 3$. The absolute minimum value of $f(x)$ is
- (a) 0 (b) -15 (c) $3 - 2\pi$ (d) none of these
- Q 9. If $x \in [-1, 1]$ then the minimum value of $f(x) = x^2 + x + 1$ is

- (a) $-3/4$ (b) -15 (c) $3 - 2\pi$ (d) none of these

Q 10. Let $f(x) = \frac{4}{3}x^3 - 4x$, $0 \leq x \leq 2$. Then the global minimum value of the function is

- (a) 0 (b) $-8/3$ (c) -4 (d) none of these

Q 11. Let $f(x) = 6 - 12x + 9x^2 - 2x^3$, $1 \leq x \leq 4$. Then the absolute maximum value of $f(x)$ in the interval is

- (a) 2 (b) 1 (c) 4 (d) none of these

Q 12. If $f(x) = \int_0^x (t^2 + 2t + 2)dt$, $2 \leq x \leq 4$, then

- (a) the maximum value of $f(x)$ is $\frac{136}{3}$ (b) the minimum value of $f(x)$ is 10

- (c) the maximum value of $f(x)$ is 26 (d) none of these

Q 13. Let the interval $I = [-1, 4]$ and $f : I \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 - 3x$. Then the range of the function is

- (a) $[2, 52]$ (b) $[-2, 2]$ (c) $[-2, 52]$ (d) none of these

Q 14. The range of the function $f(x) = |2x + 1| - 2|x - 1|$, $x \in \mathbb{R}$, is

- (a) $[-3, 3]$ (b) $[0, 6]$ (c) $\mathbb{R}$ (d) none of these

Q 15. Let the function $f(x)$ be defined as follows :

$$f(x) = x^3 + x^2 - 10x, -1 \leq x < 0$$

$$\cos x, 0 \leq x < \frac{\pi}{2}$$

$$1 + \sin x, \frac{\pi}{2} \leq x \leq \pi.$$

Then $f(x)$ is

- (a) a local minimum at $x = \pi/2$ (b) a local maximum at $x = \pi/2$
(c) an absolute minimum at $x = -1$ (d) an absolute maximum at $x = \pi$

Q 16. Let $f(x)$ be a function defined as below :

$$f(x) = \sin(x^2 - 3x), x \leq 0$$

$$6x + 5x^2, x > 0$$

Then at $x = 0$, $f(x)$

- (a) has a local maximum (b) has a local minimum
(c) is discontinuous (d) none of these

Q 17. If θ is a positive acute angle then

- (a) $\tan \theta < \theta < \sin \theta$ (b) $\theta < \sin \theta < \tan \theta$ (c) $\sin \theta < \tan \theta < \theta$ (d) none of these

Q 18. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = ax + 3 \sin x + 4 \cos x$. Then $f(x)$ is invertible if

- (a) $a \in (-5, 5)$ (b) $a \in (-\infty, -5)$ (c) $a \in (5, +\infty)$ (d) none of these

Q 19. The function $f(x) = x^3 + \lambda x^2 + 5x + \sin 2x$ will be an invertible function if λ belongs to

- (a) $(-\infty, -3)$ (b) $(-3, 3)$ (c) $(3, +\infty)$ (d) none of these

Q 20. The value of c in Lagrange's theorem for the function $|x|$ in the interval $[-1, 1]$ is

- (a) 0 (b) $1/2$ (c) $-1/2$ (d) nonexistent in the interval

Q 21. The equation $\sin x + x \cos x = 0$ has at least one root in the interval

- (a) $(-\pi/2, 0)$ (b) $(0, \pi)$ (c) $(-\pi/2, \pi/2)$ (d) none of these

Q 22. If $4a + 2b + c = 0$ then the equation $3ax^2 + 2bx + c = 0$ has at least one real root lying between

- (a) 0 and 1 (b) 1 and 2 (c) 0 and 2 (d) none of these

Q 23. If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$, $a_1 \neq 0$, $n \geq 2$, has a positive root α then the equation $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root which is

- (a) greater than α (b) smaller than α (c) greater than or equal to α (d) equal to α

Choose the correct options. One or more options may be correct.

Q 24. Let $h(x) = f(x) - \{f(x)\}^2 + \{f(x)\}^3$ for all real values of x . Then

- (a) h is increasing whenever $f(x)$ is increasing (b) h is increasing whenever $f'(x) < 0$
(c) h is decreasing whenever f is decreasing (d) nothing can be said in general

Q 25. If $f(x) = \sin x$, $-\pi/2 \leq x \leq \pi/2$, then

- (a) $f(x)$ is increasing in the interval $[-\pi/2, \pi/2]$ (b) $f\{f(x)\}$ is increasing in the interval $[-\pi/2, \pi/2]$
(c) $f\{f(x)\}$ is decreasing in $[-\pi/2, 0]$ and increasing in $[0, \pi/2]$
(d) $f\{f(x)\}$ is invertible in $[-\pi/2, \pi/2]$

- Q 26. Let $f(x) = 2\sin^3 x - 3\sin^2 x + 12 \sin x + 5$, $0 \leq x \leq \pi/2$. Then $f(x)$ is
- (a) decreasing in $[0, \pi/2]$ (b) increasing in $[0, \pi/2]$
- (c) increasing in $[0, \pi/4]$ and decreasing in $[\pi/4, \pi/2]$ (d) none of these
- Q 27. Let $f(x) = -2x^3 + 21x^2 - 60x + 41$. Then
- (a) $f(x)$ is m.d. in $(-\infty, 1)$ (b) $f(x)$ is m.i. in $(1, +\infty)$
- (c) $f(x) > 0$ for $x < 1$ (d) $f(x) < 0$ for $x > 1$
- Q 28. Let $f(x) = \frac{x^2 + 1}{[x]}$, $1 \leq x \leq 3.9$. $[.]$ denote the greatest integer function. Then
- (a) $f(x)$ is m.d. in $[1, 3.9]$ (b) $f(x)$ is m.i. in $[1, 3.9]$
- (c) The greatest value of $f(x)$ is $\frac{1}{3} \times 16.21$ (d) The least value of $f(x)$ is 2
- Q 29. Let $f(x) = |x^2 - 3x - 4|$, $-1 \leq x \leq 4$. Then
- (a) $f(x)$ is m.i. in $[-1, 3/2]$ (b) $f(x)$ is m.d. in $(3/3, 4]$
- (c) the maximum value of $f(x)$ is $\frac{25}{4}$ (d) the minimum value of $f(x)$ is 0
- Q 30. Let $f(x) = \phi(2 - x) + \phi(x)$ and $\phi''(x) < 0$ for $x \in [0, 2]$. Then
- (a) $f(x)$ is m.i. in $[0, 1]$ (b) $f(x)$ is m.d. in $[0, 1]$
- (c) $f(x)$ is m.i. in $[1, 2]$ (d) $f(x)$ is m.i. in $[1, 2]$
- Q 31. Let $f'(x) > 0$ and $g'(x) < 0$ for all $x \in \mathbb{R}$. Then
- (a) $f\{g(x)\} > f\{g(x + 1)\}$ (b) $f\{g(x)\} > f\{g(x - 1)\}$ (c) $g\{f(x)\} > g\{f(x + 1)\}$ (d) $g\{f(x)\} > g\{f(x - 1)\}$
- Q 32. Let $f(x) = 2x^2 - \log |x|$, $x \neq 0$. Then $f(x)$ is
- (a) m.i. in $\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, +\infty\right)$ (b) m.d. in $\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, +\infty\right)$
- (c) m.i. in $\left(-\infty, \frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$ (d) m.d. in $\left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$
- Q 33. Let $f(x) = x^3 - 6x^2 + 15x + 3$. Then
- (a) $f(x) > 0$ for all $x \in \mathbb{R}$ (b) $f(x) > f(x + 1)$ does not hold for any real x

(c) $f(x)$ is invertible

(d) $f(x)$ is a one-one function

Q 34. Let $f(x) = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x$, where a_i 's are real and $f(x) = 0$ has a positive root α_0 . Then

(a) $f'(x) = 0$ has a root α_1 such that $0 < \alpha_1 < \alpha_0$ (b) $f'(x) = 0$ has at least one real root

(c) $f''(x) = 0$ has at least two real roots (d) none of these

1a	2b	3c	4c	5a	6a	7b	8b	9a	10b
11b	12a	13c	14a	15b	16b	17d	18a	19b	20d
21b	22c	23b	24ac	25abd	26b	27ac	28cd	29abcd	30ad
31ac	32ad	33bcd	34abc						

Indefinite Integration

Choose the most appropriate option (a, b, c or d).

- Q 1. If $\phi(x) = \int \cot^4 x \, dx + \frac{1}{3} \cot^3 x - \cot x$ and $\phi\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$ then $\phi(x)$ is
- (a) $\pi - x$ (b) $x - \pi$ (c) $\frac{\pi}{2} - x$ (d) none of these
- Q 2. Integral of $f(x) = \sqrt{1+x^2}$ with respect to x^2 is
- (a) $\frac{2}{3} \frac{(1+x^2)^{3/2}}{x} + k$ (b) $\frac{2}{3} (1+x^2)^{3/2} + k$ (c) $\frac{2}{3} x(1+x^2)^{3/2} + k$ (d) none of these
- Q 3. $\int \frac{d(x^2+1)}{\sqrt{x^2+2}}$ is equal to
- (a) $2\sqrt{x^2+2} + k$ (b) $\sqrt{x^2+2} + k$ (c) $\frac{1}{(x^2+2)^{3/2}} + k$ (d) none of these
- Q 4. $\int \cos \left\{ 2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right\} dx$ is equal to
- (a) $\frac{1}{8} (x^2 - 1) + k$ (b) $\frac{1}{2} x^2 + k$ (c) $\frac{1}{2} x + k$ (d) none of these
- Q 5. $\int x^x (1 + \log x) dx$ is equal to
- (a) $x^x \log x + k$ (b) $e^{x^x} + k$ (c) $x^x + k$ (d) none of these
- Q 6. Let the equation of a curve passing through the point (0, 1) be given by $y = \int x^2 \cdot e^{x^3} dx$. If the equation of the curve is written in the form $x = f(y)$ the $f(y)$ is
- (a) $\sqrt{\log_e(3y-2)}$ (b) $\sqrt[3]{\log_e(3y-2)}$ (c) $\sqrt[3]{\log_e(2-3y)}$ (d) none of these
- Q 7. $\int \frac{x dx}{1+x^4}$ is equal to
- (a) $\tan^{-1} x^2 + k$ (b) $\frac{1}{2} \tan^{-1} x^2 + k$ (c) $\log(1+x^4) + k$ (d) none of these
- Q 8. The antiderivative of $\frac{2^x}{\sqrt{1-4^x}}$ w.r.t. x is

(a) $\log_2 e \cdot \sin^{-1}(2^x) + k$ (b) $\sin^{-1}(2^x) + k$ (c) $\cos^{-1}(2^x) \cdot \frac{1}{\log_e 2} + k$ (d) none of these

Q 9. $\int \frac{(1+x)^2}{x+x^3} dx$ is equal to

(a) $\log_e x + \log_e(1+x^2) + k$ (b) $\log_e x + \tan^{-1} x + k$
(c) $\log_e x + 2 \tan^{-1} x + k$ (d) none of these

Q 10. $\int \frac{x^{5/2}}{\sqrt{1+x^7}} dx$ is

(a) $\frac{2}{7} \log(x^{7/2} + \sqrt{x^7+1}) + c$ (b) $\frac{1}{2} \log \frac{x^7+1}{x^7-1} + c$
(c) $2\sqrt{1+x^7} + c$ (d) none of these

Q 11. $\int \frac{dx}{x^{1/5}(1+x^{4/5})^{1/2}}$ is

(a) $\sqrt{1+x^{4/5}} + k$ (b) $\frac{5}{2} \sqrt{1+x^{4/5}} + k$ (c) $x^{4/5}(1+x^{4/5})^{1/2} + k$ (d) none of these

Q 12. The primitive of the function $x |\cos x|$ when $\frac{\pi}{2} < x < \pi$ is given by

(a) $\cos x + x \sin x$ (b) $-\cos x - x \sin x$ (c) $x \sin x - \cos x$ (d) none of these

Q 13. $\int x \sec x^2 dx$ is equal to

(a) $\frac{1}{2} \log(\sec x^2 + \tan x^2) + k$ (b) $\frac{x^2}{2} \log(\sec x^2 + \tan x^2) + k$
(c) $2 \log(\sec x^2 + \tan x^2) + k$ (d) none of these

Q 14. $\int \frac{f(x) \cdot \phi'(x) - f'(x) \cdot \phi(x)}{f(x) \cdot \phi(x)} \{ \log \phi(x) - \log f(x) \} dx$ is equal to

(a) $\log \frac{\phi(x)}{f(x)} + k$ (b) $\frac{1}{2} \left\{ \log \frac{\phi(x)}{f(x)} \right\}^2 + k$ (c) $\frac{\phi(x)}{f(x)} \log \frac{\phi(x)}{f(x)} + k$ (d) none of these

Q 15. $\int \sin 2x \cdot \log \cos x dx$ is equal to

(a) $\cos^2 x \left(\frac{1}{2} + \log \cos x \right) + k$ (b) $\cos^2 x \cdot \log \cos x + k$

(c) $\cos^2 x \left(\frac{1}{2} - \log \cos x \right) + k$ (d) none of these

Q 16. $\int e^{-x}(1 - \tan x) \sec x \, dx$ is equal to

(a) $e^{-x} \sec x + c$ (b) $e^{-x} \tan x + c$ (c) $-e^{-x} \tan x + c$ (d) none of these

Q 17. $\int \frac{1 + \sin x}{1 + \cos x} \cdot e^x dx$ is equal to

(a) $e^x \tan\left(\frac{x}{2}\right) + k$ (b) $e^x \tan x + k$ (c) $\frac{1}{2} e^x \tan \frac{x}{2} + k$ (d) $e^x \sec^2 \frac{x}{2} + k$

Q 18. Let $\int e^x \{f(x) - f'(x)\} dx = \phi(x)$. Then $\int e^x f(x) dx$ is

(a) $\phi(x) + e^x f(x)$ (b) $\phi(x) - e^x f(x)$ (c) $\frac{1}{2} \{\phi(x) + e^x f(x)\}$ (d) $\frac{1}{2} \{\phi(x) + e^x f'(x)\}$

Q 19. If $f(0) = f'(0) = 0$ and $f''(x) = \tan^2 x$ then $f(x)$ is

(a) $\log \sec x - \frac{1}{2} x^2$ (b) $\log \cos x + \frac{1}{2} x^2$ (c) $\log \sec x + \frac{1}{2} x^2$ (d) none of these

Q 20. Let $f(x) = \int \frac{x^2 dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and $f(0) = 0$. Then $f(1)$ is

(a) $\log(1+\sqrt{2})$ (b) $\log(1+\sqrt{2}) - \frac{\pi}{4}$ (c) $\log(1+\sqrt{2}) + \frac{\pi}{4}$ (d) none of these

Q 21. $\int \frac{dx}{\cos x + \sqrt{3} \sin x}$ is equal to

(a) $\log \tan\left(\frac{x}{2} + \frac{\pi}{3}\right) + k$ (b) $\log \tan\left(\frac{x}{2} - \frac{\pi}{3}\right) + k$ (c) $\frac{1}{2} \log \tan\left(\frac{x}{2} + \frac{\pi}{3}\right) + k$ (d) none of these

Choose the correct options. One or more options may be correct.

Q 22. If $\int \tan^4 x \, dx = a \tan^3 x + b \tan x + \phi(x)$ then

(a) $a = \frac{1}{3}$ (b) $b = 1$ (c) $\phi(x) = x + c$ (d) $b = -1$

Q 23. If $\int \frac{\sin x}{\sin(x-\alpha)} dx = Ax + B \log \sin(x-\alpha) + C$ then

(a) $A = \sin \alpha$ (b) $B = \cos \alpha$ (c) $A = \cos \alpha$ (d) $B = \sin \alpha$

Q 24. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + b \log_e(9e^{2x} - 4) + C$ then

- (a) $A = \frac{3}{2}$ (b) $B = \frac{35}{36}$ (c) C is indefinite (d) $A + B = -\frac{19}{36}$

Q 25. If $\int x \log(1 + x^2) dx = \phi(x) \cdot \log(1 + x^2) + \psi(x) + c$ then

- (a) $\phi(x) = \frac{1+x^2}{2}$ (b) $\psi(x) = \frac{1+x^2}{2}$ (c) $\psi(x) = -\frac{1+x^2}{2}$ (d) $\phi(x) = -\frac{1+x^2}{2}$

Q 26. $\int \frac{dx}{(x+1)(x-2)} = A \log(x+1) + B \log(x-2) + C$, where

- (a) $A + B = 0$ (b) $AB = -1$ (c) $A : B = -1$ (d) none of these

1d 2b 3a 4b 5c 6b 7b 8a 9c 10a

11b 12b 13a 14b 15c 16d 17a 18c 19a 20b

21c 22acd 23cd 24bcd 25ac 26ac

Definite Integration

Choose the most appropriate option (a, b, c or d).

Q 1. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{nr}}$ is equal to

- (a) 2 (b) 1 (c) 0 (d) none of these

Q 2. $\lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$ is

- (a) π (b) $\pi/2$ (c) $\pi/4$ (d) none of these

Q 3. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} e^{r/n}$ is

- (a) e (b) $e - 1$ (c) $1 - e$ (d) $e + 1$

Q 4. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \sin \frac{r\pi}{2n}$ is

- (a) $\pi/2$ (b) 2 (c) $2/\pi$ (d) none of these

Q 5. $\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{1}{n+r}$ is

- (a) $\log_e 5$ (b) 0 (c) $\log_e 4$ (d) none of these

Q 6. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}}$ equals

- (a) $1 + \sqrt{5}$ (b) $-1 + \sqrt{5}$ (c) $-1 + \sqrt{2}$ (d) $1 + \sqrt{2}$

Q 7. $\lim_{n \rightarrow \infty} \left\{ \frac{n!}{(kn)^n} \right\}^{1/n}$, where $k \neq 0$ is a constant and $n \in \mathbb{N}$, is equal to

- (a) ke (b) $k^{-1} \cdot e$ (c) ke^{-1} (d) $k^{-1} \cdot e^{-1}$

Q 8. $\lim_{n \rightarrow \infty} \frac{2^k + 4^k + 6^k + \dots + (2n)^k}{n^{k+1}}$ $k \neq -1$, is equal to

- (a) 2^k (b) $\frac{2^k}{k+1}$ (c) $\frac{1}{k+1}$ (d) none of these

Q 9. Let (a, b) and (λ, μ) be two points on the curve $y = f(x)$. If the slope of the tangent to the curve at (x, y) be $\phi(x)$ then $\int_a^\lambda \phi(x) dx$ is

- (a) $\lambda - a$ (b) $\mu - b$ (c) $\lambda + \mu - a - b$ (d) none of these

Q 10. If $g(x) = \int_0^x \cos^4 t \, dt$ then $g(x + \pi)$ equals

- (a) $g(x) + g(\pi)$ (b) $g(x) - g(\pi)$ (c) $g(x) g(\pi)$ (d) $g(x)/g(\pi)$

Q 11. If $af(x) + bf\left(\frac{1}{x}\right) = \frac{1}{x} - 5$, $x \neq 0$, $a \neq b$, then $\int_1^2 f(x) dx$ equals

- (a) $\frac{(\log 2 - 5)a + \frac{13}{2}b}{a^2 - b^2}$ (b) $\frac{(\log 2 - 5)a + \frac{7b}{2}}{a^2 - b^2}$ (c) $\frac{(5 - \log 2)a + \frac{7b}{2}}{a^2 - b^2}$ (d) none of these

Q 12. $\int_0^2 \sqrt{\frac{2+x}{2-x}} dx$ is equal to

- (a) $\pi + 1$ (b) $1 + \pi/2$ (c) $\pi + 3/2$ (d) none of these

Q 13. The value of $\int_0^\pi \sec^2 x \, dx$ is

- (a) 0 (b) 2 (c) 1 (d) none of these

Q 14. If $a_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin x} dx$ then $a_2 - a_1$, $a_3 - a_2$, $a_4 - a_3$, are in

- (a) AP (b) GP (c) HP (d) none of these

Q 15. Let $a_n = \int_0^{\pi/2} \tan^n x \, dx$. Then $a_2 + a_4$, $a_3 + a_5$, $a_4 + a_6$ are in

- (a) AP (b) GP (c) HP (d) none of these

Q 16. $\int_1^{e^{37}} \frac{\pi \sin(\pi \log_e x)}{x} dx$ is equal to

- (a) 2 (b) -2 (c) $2/\pi$ (d) 2π

Q 17. Let $\frac{d}{dx} F(x) \frac{e^{\sin x}}{x}$, $x > 0$. If $\int_1^4 \frac{2e^{\sin x^2}}{x} dx = F(x) - F(1)$ then one of the possible values of k is

- (a) 4 (b) -4 (c) 16 (d) none of these

Q 18. If $f(x) = \int_{-1}^1 \frac{\sin x}{1+t^2} dt$ then $f'\left(\frac{\pi}{3}\right)$ is

- (a) nonexistent (b) $\frac{\pi}{4}$ (c) $\frac{\pi\sqrt{3}}{4}$ (d) none of these

Q 19. The value of $\int_{\pi^3/37}^{\pi^3/8} \sin x \, dx$, where $t = x^3$, is

- (a) $\frac{\pi^2}{6} + (3 - \sqrt{3})\pi - 3$ (b) $\cos \frac{\pi^3}{27} - \cos \frac{\pi^3}{8}$ (c) $\frac{\pi^2}{6}$ (d) none of these

Q 20. The value of $\int_1^2 [f\{g(x)\}]^{-1} \cdot f'\{g(x)\} \cdot g'(x) \, dx$, where $g(1) = g(2)$, is equal to

- (a) 1 (b) 2 (c) 0 (d) none of these

Q 21. If $f\left(\frac{1}{x}\right) + x^2 f(x) = 0$, $x > 0$, and $I = \int_{1/x}^x f(z) \, dz$, $\frac{1}{2} \leq x \leq 2$, then I is

- (a) $f(2) - f(1/2)$ (b) $f(1/2) - f(2)$ (c) 0 (d) none of these

Q 22. $\int_{\pi/4}^{3\pi/4} \frac{dx}{1 + \cos x}$ is equal to

- (a) 2 (b) -2 (c) 1/2 (d) -1/2

Q 23. If $f(x)$ satisfies the conditions of Rolle's theorem in $[1, 2]$ then $\int_1^2 f'(x) \, dx$ is equal to

- (a) 1 (b) 3 (c) 0 (d) none of these

Q 24. The value of $\int_0^{\pi/2} \sin^8 x \, dx$ is

- (a) $\frac{105\pi}{32(4!)}$ (b) $\frac{105\pi}{16(4!)}$ (c) $\frac{105}{16(4!)}$ (d) none of these

Q 25. If $f(x)$ be a quadratic polynomial such that $f(0) = 2$, $f'(0) = -3$ and $f''(0) = 4$ then $\int_{-1}^1 f(x) \, dx$ is equal to

- (a) -3 (b) 16/3 (c) 0 (d) none of these

Choose the correct options. One or more options may be correct.

Q 26. Let $I_n = \int_0^{\pi/2} \cos^n x \, dx$, $n \in \mathbb{N}$. Then

- (a) $I_{n-2} > I_n$ (b) $n(I_{n-2} - I_n) = I_{n-2}$ (c) $I_n : I_{n-1} = n : (n-1)$ (d) none of these

Q 27. Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$, $n \in \mathbb{N}$. Then

- (a) $I_1 = I_3 + 2I_5$ (b) $I_n + I_{n-2} = \frac{1}{n}$ (c) $I_n + I_{n-2} = \frac{1}{n-1}$ (d) none of these

1a	2b	3b	4c	5a	6b	7d	8b	9b	10a
11b	12d	13d	14c	15c	16a	17c	18b	19a	20c
21c	22a	23c	24a	25b	26ab	27ac			

Properties and Applications of Definite Integrals

Choose the most appropriate option (a, b, c or d).

- Q 1. $\int_0^{\pi/2} \frac{f(x)}{f(x) + f\left(\frac{\pi}{2} - x\right)} dx$, where $f(x) \neq -f\left(\frac{\pi}{2} - x\right)$ for $0 \leq x \leq \frac{\pi}{2}$, has the value
- (a) $f(0)$ (b) $f\left(\frac{\pi}{2}\right)$ (c) $\frac{\pi}{2}$ (d) none of these
- Q 2. $\int_0^a \{f(x) + f(-x)\} dx$ is equal to
- (a) $2 \int_0^a f(x) dx$ (b) $\int_{-a}^a f(x) dx$ (c) 0 (d) $-\int_{-a}^a f(-x) dx$
- Q 3. $\int_{\pi/5}^{3\pi/10} \frac{\cos x}{\cos x + \sin x} dx$ is equal to
- (a) π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) none of these
- Q 4. The equation $\int_{-\pi/4}^{\pi/4} \left(\lambda |\sin x| + \frac{\mu \sin x}{1 + \cos x} + \nu \right) dx = 0$, where λ, μ, ν are constants gives a relation between
- (a) λ, μ and ν (b) λ and ν (c) λ and μ (d) μ and ν
- Q 5. Let $f(x) = x - [x]$ for $x \in \mathbb{R}$, where $[x]$ = the greatest integer $\leq x$. Then $\int_{-2}^2 f(x) dx$ is
- (a) 4 (b) 2 (c) 0 (d) 1
- Q 6. $\int_0^{\pi} \sin^6 x \cdot \cos^5 x dx$ is equal to
- (a) $2 \int_0^{\pi/2} \sin^{50} x \cdot \cos^{47} x dx$ (b) 0 (c) 1 (d) none of these
- Q 7. $\int_{-2}^2 |1 - x^2| dx$ is
- (a) 4 (b) 2 (c) -2 (d) 0
- Q 8. $\int_0^{2\pi} \frac{x \sin^{2n} x}{\sin^{2n} + \cos^{2n} x} dx$, $n > 0$, is equal to
- (a) π (b) 2π (c) π^2 (d) $\frac{1}{2} \pi^2$

- Q 9. $\int_{-\pi/4}^{\pi/4} \frac{e^x \cdot \sec^2 x dx}{e^{2x} - 1}$ is equal to
- (a) 0 (b) 2 (c) e (d) none of these
- Q 10. If $[x]$ denotes the greatest integer less than or equal to x then $\int_0^\infty \left[\frac{2}{e^x} \right] dx$ is equal to
- (a) $\log_e 2$ (b) e^2 (c) 0 (d) $\frac{2}{e}$
- Q 11. $\int_1^3 |(2-x)\log_e x| dx$ is equal to
- (a) $\frac{3}{2} \log_e 3 + \frac{1}{2}$ (b) $\log_e \frac{16}{3\sqrt{2}} - \frac{1}{2}$ (c) $-\frac{3}{2} \log_e 3 - \frac{1}{2}$ (d) none of these
- Q 12. $\int_{-2}^2 |x(x-1)| dx$ is
- (a) $\frac{11}{3}$ (b) $\frac{13}{3}$ (c) $\frac{16}{3}$ (d) $\frac{17}{3}$
- Q 13. The value of $\int_{-2}^2 \frac{\sin^2 x}{\left[\frac{x}{\pi} \right] + \frac{1}{2}} dx$, where $[x]$ = the greatest integer greater than or equal to x , is
- (a) 1 (b) 0 (c) $4 - \sin 4$ (d) none of these
- Q 14. The value of $\int_0^\pi [\cos x] dx$, where $[.]$ is the greatest integer function, is
- (a) $\frac{\pi}{2}$ (b) 0 (c) π (d) $-\frac{\pi}{2}$
- Q 15. Let $a_n = \int_0^{\pi/2} \cos^n x \cdot \cos nx dx$. Then $a_n : a_{n+1}$ is equal to
- (a) 3 : 1 (b) 2 : 3 (c) 2 : 1 (d) 3 : 4
- Q 16. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$ then the value of $f(1)$ is
- (a) $\frac{1}{2}$ (b) 0 (c) 1 (d) $-\frac{1}{2}$
- Q 17. The value of $\int_{-1}^1 \max\{2-x, 2, 1+x\} dx$ is

- (a) 4 (b) $\frac{9}{2}$ (c) 2 (d) none of these

Q 18. Let f be a positive function. If

$$I_1 = \int_{1-k}^k x f\{x(1-x)\} dx, I_2 = \int_{1-k}^k f\{x(1-x)\} dx,$$

where $2k - 1 > 0$, then $\frac{I_1}{I_2}$ is

- (a) 2 (b) k (c) $\frac{1}{2}$ (d) 1

Q 19. If $x \in (2n\pi, 2n\pi + \pi)$ then $\int_0^x [\sin x] dx$, where $[x]$ = greatest integer less than or equal to x , is equal to

- (a) $-\pi$ (b) $-n\pi$ (c) 0 (d) none of these

Q 20. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{\sin^3 x + \sin x}$ is

- (a) 0 (b) 2 (c) 1 (d) none of these

Q 21. $\int_0^\pi \frac{dx}{1+3^{\cos x}}$ is equal to

- (a) π (b) 0 (c) $\frac{\pi}{2}$ (d) none of these

Q 22. If $[y]$ = the greatest integer less than or equal to y then $\int_{\pi/2}^{3\pi/2} [2 \sin x] dx$ is

- (a) $-\pi$ (b) 0 (c) $-\frac{\pi}{2}$ (d) $\frac{\pi}{2}$

Q 23. The value of $\int_0^{2\pi} \frac{dx}{e^{\sin x} + 1}$ is

- (a) π (b) 0 (c) 2π (d) $\frac{\pi}{2}$

Q 24. $\int_{a/4}^{3a/4} \frac{\sqrt{x}}{\sqrt{a-x} + \sqrt{x}} dx$ is equal to

- (a) $\frac{a}{2}$ (b) a (c) $-a$ (d) none of these

Q 25. $\int_0^{\pi/4} \sin x d(x - [x])$ is equal to

- (a) $\frac{1}{2}$ (b) $1 - \frac{1}{\sqrt{2}}$ (c) 1 (d) none of these

Q 26. If $f(x) = \int_0^{\sin x} \cos^{-1} t dt + \int_0^{\cos x} \sin^{-1} t dt$, $0 < x < \frac{\pi}{2}$, then $f\left(\frac{\pi}{4}\right)$ is

- (a) $\frac{\pi}{\sqrt{2}}$ (b) $1 + \frac{\pi}{2\sqrt{2}}$ (c) 1 (d) none of these

Q 27. The value of $\int_{\alpha}^{\beta} x |x| dx$, where $\alpha < 0 < \beta$, is

- (a) $\frac{1}{2}(\alpha^2 + \beta^2)$ (b) $\frac{1}{3}(\beta^2 - \alpha^2)$ (c) $\frac{1}{3}(\alpha^2 + \beta^2)$ (d) none of these

Q 28. The value of $\int_0^{\pi/4} \log(1 + \tan x) dx$ is equal to

- (a) $\frac{\pi}{8} \log_e 2$ (b) $\frac{\pi}{4} \log_e 2$ (c) $\frac{\pi}{4}$ (d) none of these

Q 29. $\int_{-1}^1 (x - [2x]) dx$ is equal to

- (a) 1 (b) 0 (c) 2 (d) 4

Q 30. If $f(x) = |x| + 1$, $-1 \leq x < 0$

$$1 + |x|^2, 0 \leq x \leq 1$$

then $\int_{-1}^1 f(x) dx$ is equal to

- (a) $-\frac{1}{6}$ (b) $\frac{17}{6}$ (c) $-\frac{17}{6}$ (d) none of these

Q 31. Let $f(x) = \text{maximum}\{x + |x|, -[x]\}$, where $[x]$ = the greatest integer $\leq x$. Then $\int_{-2}^2 f(x) dx$ is equal to

- (a) 3 (b) 2 (c) 1 (d) none of these

Q 32. $\int_0^3 |x^3 - 3x^2 + 2x| dx$ is equal to

- (a) $\frac{3}{4}$ (b) $\frac{7}{4}$ (c) $\frac{11}{4}$ (d) none of these

- Q 33. Let $f(x)$ be a continuous function such that $f(a-x) + f(x) = 0$ for all $x \in [0, a]$. Then $\int_0^a \frac{dx}{1+e^{f(x)}}$ is equal to
- (a) a (b) $\frac{a}{2}$ (c) $f(a)$ (d) $\frac{1}{2}f(a)$
- Q 34. $\int_{-a}^a \log_e(x + \sqrt{1+x^2}) dx$ is equal to
- (a) $2\log_e a$ (b) 0 (c) $\log_e 2 + \log a$ (d) none of these
- Q 35. Let $f(x) = \frac{e^x + 1}{e^x - 1}$ and $\int_0^1 \frac{e^x + 1}{e^x - 1} \cdot x dx = \lambda$. Then $\int_{-1}^1 xf(t) dt$ is equal to
- (a) 0 (b) 2λ (c) λ (d) none of these
- Q 36. Let $f(x)$ be a continuous function in $\mathbb{R}$ such that $f(x) + f(y) = f(x+y)$. If $\int_0^3 f(x) dx = k$ then $\int_{-3}^3 f(x) dx$ is equal to
- (a) $2k$ (b) 0 (c) $k/2$ (d) $-2k$
- Q 37. Let $f(x)$ be a continuous function such that $f(x)$ does not vanish for all $x \in \mathbb{R}$. If $\int_2^3 f(x) dx = \int_{-2}^3 f(x) dx$ then $f(x)$, $x \in \mathbb{R}$, is
- (a) an even function (b) an odd function (c) a periodic function (d) none of these
- Q 38. Let $I = \int_{-a}^a (p \tan^3 x + q \cos^2 x + r \sin x) dx$, where p, q, r are arbitrary constants. The numerical value of I depends on
- (a) p, q, r (b) q, r, a (c) q, a (d) p, r, a
- Q 39. Let $I_1 = \int_0^1 e^{-x^2} dx$, $I_2 = \int_0^1 e^{-x} \cos^2 x dx$ and $I_3 = \int_0^1 e^{-x^2} \cos^2 x dx$. Then
- (a) $I_1 < I_2 < I_3$ (b) $I_3 < I_2 < I_1$ (c) $I_2 < I_1 < I_3$ (d) $I_2 < I_3 < I_1$
- Q 40. $\int_0^{100\pi} \sqrt{1 + \cos 2x} dx$ is equal to
- (a) 0 (b) $100\sqrt{2}$ (c) $200\sqrt{2}$ (d) 100
- Q 41. The value of $\int_0^1 (1 + e^{-x^2}) dx$ is
- (a) -1 (b) 2 (c) $1 + e^{-1}$ (d) none of these

- Q 42. If $\int_{-2}^3 f(x)dx = 5$ and $\int_1^3 \{2 - f(x)\}dx = 6$ then the value of $\int_{-2}^1 f(x)dx$ is
- (a) 7 (b) 3 (c) -7 (d) -3
- Q 43. Let $f(x)$ be a continuous function such that $\int_n^{n+1} f(x)dx = n^3$, $n \in \mathbb{Z}$. Then the value of $\int_{-3}^3 f(x)dx$ is
- (a) 9 (b) -27 (c) -9 (d) none of these
- Q 44. If $I = \int_0^1 \frac{x dx}{8 + x^2}$ then the smallest interval in which I lies is
- (a) $\left(0, \frac{1}{8}\right)$ (b) $\left(0, \frac{1}{9}\right)$ (c) $\left(0, \frac{1}{10}\right)$ (d) $\left(0, \frac{1}{7}\right)$
- Q 45. If $\int_0^1 x e^{x^2} dx = \lambda \int_0^1 e^{x^2} dx$ then
- (a) $\lambda = 0$ (b) $\lambda \in (0, 1)$ (c) $\lambda \in (-\infty, 0)$ (d) $\lambda \in (1, 2)$
- Q 46. $\int_{\log 1/2}^{\log 2} \sin\left(\frac{e^x - 1}{e^x + 1}\right) dx$ is equal to
- (a) $\cos \frac{1}{3}$ (b) 0 (c) $2 \cos 2$ (d) none of these
- Q 47. Let $\int_0^a f(x)dx = \lambda$ and $\int_0^a f(2a - x)dx = \mu$. Then $\int_0^{2a} f(x)dx$ is equal to
- (a) $\lambda + \mu$ (b) $\lambda - \mu$ (c) $2\lambda - \mu$ (d) $\lambda - 2\mu$
- Q 48. If $f(-x) + f(x) = 0$ then $\int_a^x f(t)dt$ is
- (a) an odd function (b) an even function (c) a periodic function (d) none of these
- Q 49. If $f(x)$ and $g(x)$ be continuous functions over the closed interval $[0, a]$ such that $f(x) = f(a - x)$ and $g(x) + g(a - x) = 2$. Then $\int_0^a f(x).g(x)dx$ is equal to
- (a) $\int_0^a f(x)dx$ (b) $\int_0^a g(x)dx$ (c) $2a$ (d) none of these
- Q 50. If $f(x) = f(a + x)$ and $\int_0^a f(x)dx = p$ then $\int_a^{na} f(x)dx$ is equal to
- (a) np (b) $(n - 1)p$ (c) $(n + 1)p$ (d) none of these

Q 51. Let $f(x)$ be a given integrable function such that $f(x + k) = f(x)$ for all $x \in \mathbb{R}$. Then $\int_a^{a+k} f(x) dx$ depends for its value on

- (a) a only (b) k only (c) both a and k (d) neither a nor k

Q 52. The value of $\int_{\pi/4}^{3\pi/4} \frac{x}{1 + \sin x} dx$ is equal to

- (a) $(\sqrt{2} - 1)\pi$ (b) $(\sqrt{2} + 1)\pi$ (c) π (d) none of these

Q 53. The value of $\int_0^{\pi/2} \frac{dx}{1 + \tan^3 x}$ is

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) π (d) none of these

Q 54. Let f and g be two continuous functions. Then

$$\int_{-\pi/2}^{\pi/2} \{f(x) + f(-x)\}\{g(x) - g(-x)\} dx$$

is equal to

- (a) π (b) 1 (c) -1 (d) 0

Q 55. Let $\int_a^b f(x) dx = p$ and $\int_a^b |f(x)| dx = q$. Then

- (a) $|p| \leq q$ (b) $p > q$ (c) $p + q = 0$ (d) none of these

Q 56. If $f(x) = \int_0^x \log(1 + t^2) dt$ then the value of $f''(1)$ is equal to

- (a) 2 (b) 0 (c) 1 (d) none of these

Q 57. If $f(x) = \int_x^{x^2} \frac{dt}{1+t^3}$ then $f'(2)$ is equal to

- (a) $\frac{101}{585}$ (b) $\frac{-29}{585}$ (c) $\frac{-56}{585}$ (d) none of these

Q 58. If $f(2a - x) = f(x)$ and $\int_0^a f(x) dx = \lambda$ then $\int_0^{2a} f(x) dx$ is

- (a) 2λ (b) λ (c) 0 (d) none of these

Q 59. Let $f(x)$ be a continuous function such that the area bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = a$ is $1 + \frac{a^2}{2} \sin a$. Then

$$(a) f\left(\frac{\pi}{2}\right) = 1 + \frac{\pi^2}{8}$$

$$(b) f(a) = 1 + \frac{a^2}{2} \sin a$$

$$(c) f(a) = a \sin a + \frac{1}{2} a^2 \cos a$$

(d) none of these

Q 60. If $y = \int_0^x \sqrt{\sin x} dx$ then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$ is

(a) 0

(b) 1

(c) -1

(d) none of these

Q 61. If $y = \int_x^{x^2} \sqrt{5-t^2} dt$ then the value of $\frac{dy}{dx}$ at $x = \sqrt{2}$ is

$$(a) 1 - \sqrt{3}$$

$$(b) \sqrt{3}(2\sqrt{6} - 1)$$

$$(c) 2\sqrt{2} - \sqrt{3}$$

(d) none of these

Q 62. The function $f(x) = \int_{-1}^x t(e^t - 1)(t-2)^3(t-3)^5 dt$ has a local minimum at x which is equal to

(a) 0

(b) 1

(c) 2

(d) 3

Q 63. Let $f(x)$ be a differentiable function and $f(1) = 2$. If $\lim_{x \rightarrow 1} \int_2^{f(x)} \frac{2t}{x-1} dt = 4$ then the value of $f'(1)$ is

(a) 1

(b) 2

(c) 4

(d) none of these

Q 64. If $\phi(x) = \int_x^{x^2} (t-1)dt$, $1 \leq x \leq 2$, then the greatest value of $\phi(x)$ is

(a) 2

(b) 4

(c) 8

(d) none of these

Q 65. If $\int_0^1 (1 + \sin^4 x)(ax^2 + bx + c)dx = \int_0^2 (1 + \sin^4 x)(ax^2 + bx + c)dx$ then the quadratic equation $ax^2 + bx + c = 0$ has

(a) at least one root in $(1, 2)$

(b) no root in $(1, 2)$

(c) two equal roots in $(1, 2)$

(d) both roots imaginary

Q 66. Let $f(x)$ be a function defined by $f(x) = \int_1^x x(x^2 - 3x + 2)dx$, $1 \leq x \leq 3$. Then the range of $f(x)$ is

(a) $[0, 2]$

(b) $\left[-\frac{1}{4}, 4\right]$

(c) $\left[-\frac{1}{4}, 2\right]$

(d) none of these

Choose the correct options. One or more options may be correct.

Q 67. If $A = \int_0^\pi \frac{\sin x}{\sin x + \cos x} dx$, $B = \int_0^\pi \frac{\sin x}{\sin x - \cos x} dx$ then

- (a) $A + B = 0$ (b) $A = B$ (c) $A = B = \pi/2$ (d) $A = -B = \pi$

Q 68. Let $f(a) > 0$, and let $f(x)$ be a nondecreasing continuous function in $[a, b]$.

Then $\frac{1}{b-a} \int_a^b f(x) dx$ has the

- (a) maximum value of $f(b)$ (b) minimum value $f(a)$
(c) maximum value $bf(b)$ (d) minimum value $\frac{f(a)}{b-a}$

Q 69. The value of $\int_0^\pi \frac{\sin nx}{\sin x} dx$, $n \in \mathbb{N}$, is

- (a) π if n is even (b) 0 if n is odd (c) 0 if n is even (d) π for all $n \in \mathbb{N}$

Q 70. If $f(x) = \int_x^{x^2} \frac{dt}{(\log t)^2}$, $x \neq 0$, $x \neq 1$, then $f(x)$ is

- (a) monotonically increasing in $(2, +\infty)$ (b) monotonically increasing in $(1, 2)$
(c) monotonically increasing in $(2, +\infty)$ (d) monotonically decreasing in $(0, 1)$

Q 71. Let $f(x) = ax^3 + bx^2 + cx$ have relative extrema $x = 1$ and at $x = 5$. If $\int_{-1}^1 f(x) dx = 6$ then

- (a) $a = -1$ (b) $b = 9$ (c) $c = 15$ (d) $a = 1$

Q 72. Let $f(x) = \int_0^x |x-1| dx$, $x \geq 0$. Then $f'(x)$ is

- (a) continuous at $x = 1$ (b) continuous at $x = 2$
(c) differentiable at $x = 1$ (d) differentiable at $x = 2$

1d	2b	3d	4b	5b	6b	7a	8c	9a	10a
11b	12d	13b	14d	15c	16a	17b	18c	19b	20a
21c	22c	23a	24a	25b	26b	27c	28a	29a	30b
31d	32c	33b	34b	35b	36b	37d	38c	39d	40c
41d	42a	43b	44b	45b	46b	47a	48b	49a	50b
51b	52a	53b	54d	55a	56c	57b	58a	59c	60b
61c	62d	63a	64b	65a	66c	67bc	68ab	69c	70ad

71ab 72abd

Area

Choose the most appropriate option (a, b, c or d).

- Q 1. $P(\alpha, f(\alpha))$ and $Q(\beta, f(\beta))$ are ends of an arc in the first quadrant. The area bounded by the arc, ordinates through P and Q, and the x-axis is
- (a) $\int_{f(\alpha)}^{f(\beta)} f^{-1}(y)dy$ (b) $\int_{\alpha}^{\beta} f^{-1}(y)dy$ (c) $\int_{\alpha}^{\beta} f(x)dx$ (d) $\int_{f(\alpha)}^{f(\beta)} f(x)dx$
- Q 2. The area bounded by the lines $y = |x - 2|$, $|x| = 3$ and $y = 0$ is
- (a) 13 unit² (b) 5 unit² (c) 9 unit² (d) 7 unit²
- Q 3. The area bounded by the curve $y = \sqrt{4 - x^2}$ and the line $y = 0$ is
- (a) 4π (b) 2π (c) π (d) $\frac{\pi}{2}$
- Q 4. The area bounded by the curve $y^2 = 4x$ and the double ordinate $x = 2$, is
- (a) $\frac{4}{3}\sqrt{2}$ unit² (b) $\frac{8}{3}\sqrt{2}$ unit² (c) $\frac{16}{3}\sqrt{2}$ unit² (d) $\frac{32}{3}\sqrt{2}$ unit²
- Q 5. The area bounded by the curve $x^2 = ky$, $k > 0$ and the line $y = 3$ is 12 unit². Then k is
- (a) 3 (b) $3\sqrt{3}$ (c) $\frac{3}{4}$ (d) none of these
- Q 6. The area bounded by the curve $y = 2^x$, the x-axis and the y-axis is
- (a) $\log_e 2$ (b) $\log_e 4$ (c) $\log_4 e$ (d) $\log_2 e$
- Q 7. The area of the portion enclosed by the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ and the axes of reference is
- (a) $\frac{a^2}{6}$ (b) a^2 (c) $\frac{a^2}{2}$ (d) $\frac{a^2}{4}$
- Q 8. The area bounded by the curve $x = \cos^{-1}y$ and the lines $|x| = 1$ is
- (a) $\sin 1$ (b) $\cos 1$ (c) $2 \sin 1$ (d) $2 \cos 1$
- Q 9. The area bounded by the curve $y = \sqrt{x}$, the line $2y + 3 = x$ and the x-axis in the first quadrant is
- (a) 9 (b) $\frac{27}{4}$ (c) 36 (d) 18
- Q 10. The area of the region bounded by the pairs of lines $y = |x - 1|$ and $y = 3 - |x|$ is

- (a) 3 unit² (b) 4 unit² (c) 6 unit² (d) 2 unit²

Q 11. The ratio in which the area bounded by the curves $y^2 = 4x$ and $x^2 = 4y$ is divided by the line $x = 1$ is

- (a) 64 : 49 (b) 15 : 34 (c) 15 : 49 (d) none of these

Q 12. Let $f(x)$ be a continuous function such that the area bounded by the curve $y = f(x)$, the x-axis and the two ordinates $x = 0$ and $x = a$ is $\frac{a^2}{2} + \frac{a}{2}\sin a + \frac{\pi}{2}\cos a$. The $f\left(\frac{\pi}{2}\right)$ is

- (a) $\frac{1}{2}$ (b) $\frac{\pi^2}{8} + \frac{\pi}{4}$ (c) $\frac{\pi+1}{2}$ (d) $\frac{\pi}{2}$

Q 13. The area bounded by the curve $f(x) = ce^x$ ($c > 0$), the x-axis and the two ordinates $x = p$ and $x = q$ is proportional to

- (a) $f(p) \cdot f(q)$ (b) $|f(p) - f(q)|$ (c) $f(p) + f(q)$ (d) $\sqrt{f(p)f(q)}$

Q 14. The ordinate $x = a$ divides the area bounded by the x-axis, the curve $y = 1 + \frac{8}{x^2}$ and the ordinates $x = 2$ and $x = 4$ into equal parts. Then a is

- (a) 3 (b) $\frac{7}{2}$ (c) $2\sqrt{2}$ (d) $\frac{5}{2}$

Q 15. The area of the ellipse $\frac{(x+1)^2}{4} + y^2 = 1$ falling in the first quadrant is

- (a) $\frac{1}{6}(4\pi - 3\sqrt{3})$ (b) $\frac{1}{12}(4\pi - 3\sqrt{3})$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{3}(\pi - \sqrt{3})$

1c 2a 3b 4c 5a 6d 7a 8c 9a 10b

11c 12a 13b 14c 15b

Ordinary Differential Equations

Choose the most appropriate option (a, b, c or d).

- Q 1. The order the differential equation $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 1$ is
- (a) one (b) two (c) four (d) zero
- Q 2. The degree of the differential equation $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x^2$ is
- (a) one (b) two (c) half (d) four
- Q 3. The differential equation of the family of curves $y = e^x(A \cos x + B \sin x)$, where A, B are arbitrary constants, has the degree n and order m. Then
- (a) $n = 2, m = 1$ (b) $n = 2, m = 2$ (c) $n = 1, m = 2$ (d) $n = 1, m = 1$
- Q 4. The general solution of a differential equation is $y = ae^{bx+c}$ where a, b, c are arbitrary constants. The order of the differential equation is
- (a) 3 (b) 2 (c) 1 (d) none of these
- Q 5. The general solution of a differential equation is $(y + c)^2 = cx$ where c is an arbitrary constant. The order and degree of the differential equation are respectively
- (a) 1, 2 (b) 2, 2 (c) 1, 1 (d) 2, 1
- Q 6. The degree and order of the differential equation of the family of all parabolas whose axis is the x-axis, are respectively
- (a) 1, 2 (b) 3, 2 (c) 2, 3 (d) 2, 1
- Q 7. The order and degree of the differential equation of the family of circles touching the x-axis at the origin, are respectively
- (a) 1, 1 (b) 1, 2 (c) 2, 1 (d) 2, 2
- Q 8. The order and degree of the differential equation of the family of ellipses having the same foci, are respectively
- (a) 1, 1 (b) 2, 1 (c) 2, 2 (d) 1, 2
- Q 9. If $y(t)$ is a solution of the equation $(1 + t) \frac{dy}{dt} - ty = 1$ and $y(0) = -1$ then $y(1)$ is

(a) $-\frac{1}{2}$

(b) $e + \frac{1}{2}$

(c) $e - \frac{1}{2}$

(d) $\frac{1}{2}$

Q 10. The solution of $(x + \log y)dy + y dx = 0$ when $y(0) = 1$ is

(a) $y(x - 1) + y \log y = 0$

(b) $y(x - 1 + \log y) + 1 = 0$

(c) $xy + y \log y + 1 = 0$

(d) none of these

Q 11. The general solution of the equation $(1 + y^2) + (x - e^{\tan^{-1} y}) \frac{dy}{dx} = 0$ is

(a) $2xe^{\tan^{-1} y} = e^{2\tan^{-1} y} + k$

(b) $xe^{\tan^{-1} y} = \tan^{-1} y + k$

(c) $xe^{2\tan^{-1} y} = e^{\tan^{-1} y} + k$

(d) $x = 2 + ke^{-\tan^{-1} y}$

Q 12. Let $\frac{df(x)}{dx} = \frac{e^{\sin x}}{x}$, $x > 0$. If $\int_1^4 \frac{3e^{\sin x^3}}{x} dx = f(k) - f(1)$ then one of the possible values of k is

(a) 16

(b) 63

(c) 64

(d) 15

Q 13. If $x \frac{dy}{dx} + y = x \cdot \frac{f(x,y)}{f'(x,y)}$ then $f(x, y)$ is equal to (k being an arbitrary constant)

(a) $ke^{x^2/2}$

(b) $ke^{y^3/2}$

(c) $ke^{xy/2}$

(d) none of these

Q 14. The differential equation $\phi(x)dy = y \{\phi'(x) - y\}dx$ is changed in the form $df(x, y) = 0$. Then $f(x, y)$ is

(a) $\frac{1}{2}\phi(x) + y$

(b) $\frac{1}{y}\phi(x) - x$

(c) $\frac{1}{y}\phi(x) + x$

(d) $\frac{\phi(x)}{y}$

Q 15. The solution of primitive integral equation $(x^2 + y^2)dy = xy \cdot dx$ is $y = y(x)$. If $y(1) = 1$ and $y(x_0) = e$ then x_0 is

(a) $\sqrt{2(e^2 - 1)}$

(b) $\sqrt{2(e^2 + 1)}$

(c) $\sqrt{3}e$

(d) $\sqrt{\frac{1}{2}(e^2 + 1)}$

1b 2b 3c 4b 5a 6a 7a 8d 9a 10b

11a 12c 13a 14b 15c